Evidence Coverage and the Cost of False Isolation: Verifying Malicious Beacon Nodes in Mobile Ad Hoc Networks

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Abstract

Malicious beacon nodes in mobile ad hoc networks corrupt the position estimates of nodes that never exchange traffic with them, and existing defences remove a suspected beacon permanently. This paper studies the decision that precedes removal and what an error costs. A verification pipeline fuses behavioural and localisation evidence into per-observer trust and a three-state verdict, and is evaluated in ns-3 under six isolation policies and against two re-implemented literature baselines, in dense, sparse, shadowed and 50-node networks with up to two simultaneous attackers. Four findings emerge. First, every detector restricted to one evidence class was blind to the attack outside it, while collecting both classes confirmed every attacker in the dense networks. Second, we introduce cumulative false isolation Φ, in node-seconds: isolating only on confirmation withdrew about 4–5% of honest-beacon service against about 75% for immediate isolation, and the gap held when the schedules were enforced in the simulator. Third, releasing a beacon when its verdict clears re-admitted forwarding attackers in a closed loop, because isolation silences their behavioural evidence; releasing only when that evidence returns prevented every re-admission. Fourth, we derive when weighted Stouffer fusion can confirm what neither branch confirms alone. The condition explains why, in a small dense network, a disjunction of the two baselines matched the fused pipeline, whereas in sparse and larger networks it withdrew several times as much honest-beacon service. Evidence coverage and the persistence of errors, rather than headline accuracy, should therefore guide the evaluation of beacon verification.

Keywords Mobile ad hoc networks · Beacon nodes · Trust management · Evidence fusion · Node isolation · Localisation security

1 Introduction

Mobile ad hoc networks (MANETs) operate without fixed infrastructure. Nodes organise themselves, forward one another's traffic, and reconfigure as the topology changes. The properties that make this attractive for disaster response, defence communication and location-aware services are the same ones that make the network hard to secure. There is no central authority to consult, no stable membership, and no reliable way to tell a node that is failing from a node that is lying.

Beacon nodes deserve separate treatment within this problem. A beacon advertises position information that neighbouring nodes consume in order to localise themselves. A compromised beacon therefore corrupts the position estimates of nodes it never exchanged data traffic with, and the corruption propagates through the localisation computation rather than through the routing fabric. The consequences differ in kind from ordinary packet-dropping attacks, not merely in degree. A node that drops packets degrades a service the network can measure; a beacon that lies about its position degrades a service the network has no independent way to check.

Elimination of malicious beacons from localisation evidence is well studied for static beacons. Liu, Ning and Du detect a beacon whose RSSI-implied distance disagrees with the distance implied by its declared position, and revoke it once enough alerts accumulate [1]. Kuriakose et al. [2] address the same problem in EMBN-MANET, a UWB-based mobile ad hoc network, using RSSI and trilateration residuals, and report complete identification when fewer than four malicious beacons share a neighbourhood. Both establish that detection from localisation evidence is tractable. What neither settles is the decision that precedes removal, and what happens to the network when that decision is wrong: revocation in both is permanent.

Both questions matter for a practical reason. In a mobile wireless environment, congestion, transient link failure, interference and mobility all produce observations that resemble attack. A verification stage separating detection from confirmation therefore has a plausible role, a need also identified in work on trust change and revision [3] and on verification supported by observable evidence [4, 5]. And because a removed beacon is a beacon whose localisation service is lost to its neighbours, removal is not free: a verification mechanism that is too eager degrades exactly the service it exists to protect.

1.1 Research questions

The study asks four questions, each answered by a part of Section 6.

- **RQ1, coverage.** Can a detector that uses one class of evidence confirm beacons whose attack leaves no signature in that class, and does collecting both behavioural and localisation evidence close the gap? (Sections 6.1, 6.6, 6.14 and 6.15)
- **RQ2, cost of error.** How much honest-beacon service do isolation policies withdraw, and how does waiting for confirmation change that cost? (Sections 6.8 and 6.10)
- **RQ3, release.** Can an isolated beacon be released safely once the evidence against it clears? (Section 6.11)
- **RQ4, fusion.** When can fused evidence confirm an attacker that neither branch confirms alone, and is a fused detector better than classical single-evidence detectors run side by side? (Sections 6.3, 6.5, 6.6, 6.14 and 6.15)

1.2 Contributions

We answer these questions in ns-3 across 780 simulation runs: a 30-seed detection campaign over three attack scenarios and a clean network, six isolation policies replayed over that campaign and checked by enforcing them in the simulator, an iterated closed loop, two attack-intensity sweeps, log-normal shadowing, a sparse topology, a 50-node network with two simultaneous attackers, and two re-implemented literature baselines evaluated on the same traces.

1. **Evidence coverage (RQ1).** We frame malicious-beacon verification as a question of evidence coverage rather than detection accuracy. Every detector restricted to one evidence class, including re-implemented localisation-only and behaviour-only baselines from the literature, was blind to the attack outside that class, in the small, sparse and 50-node networks alike; collecting both classes confirmed every attacker in the dense networks. The verification pipeline that fuses the two classes into per-observer trust and a three-state verdict is built from established components and serves as the vehicle for the study.
2. **A measure of what false isolation costs (RQ2).** We introduce cumulative false isolation Φ, in node-seconds, and its normalised form $\bar{\Phi}$, as complements to the false-positive rate, which says nothing about how long an error persists. Isolating only on confirmation withdrew 3.9–5.0% of honest-beacon service on a clean network against 74.8% for the strongest immediate-isolation policy, and the gap held when the schedules were enforced in the simulator, under shadowing and at 50 nodes.
3. **Release is as consequential as isolation (RQ3).** Releasing an isolated beacon when its verdict clears re-admits forwarding attackers in a closed loop, because

isolation removes the very traffic their behavioural evidence comes from. An evidence-aware release rule prevented every re-admission at a small cost in false isolation, and showed that the advantage of confirmation-gated isolation comes from the gating rather than from release.

4. **When fusion helps (RQ4).** Every weighted Stouffer detector has a fusion-only region, containing attackers whose two anomalies are each sub-threshold and of comparable size, and an attack family scaled in intensity passes through it only if the branches' detection thresholds for that family lie within a factor of $1 + \sqrt{2} \approx 2.414$. The condition is checkable before any fusion experiment and predicted the outcome of two intensity sweeps. We also make explicit, for trust fusion, the calibration requirement on which any such test depends [6, 7].

5. **An honest comparison with classical detectors (RQ4).** In the small dense network a calibrated disjunction of the two literature baselines matched the fused pipeline's coverage and false isolation and confirmed attackers as fast or faster, as the fusion condition predicts. In the sparse and 50-node networks the same disjunction withdrew 11–24% of honest-beacon service on a clean network against 0–3% for the pipeline, so the benefit of combining evidence appears in Φ rather than in detection speed. In the sparse network that benefit came with a cost: the pipeline confirmed the forwarding attacker in only 4 of 10 seeds.

6. **Bounding results.** Localisation error shows that the two attack classes are disjoint in both evidence directions; detection and the false-isolation result survive 4 and 8 dB shadowing at a cost in latency for forwarding attacks; and we report negative results in the body of the paper: the trust aggregation stage changed no verdict, and neither isolation policy produced a throughput advantage surviving its confidence interval.

1.3 Organisation

Section 2 positions the work against the closest prior art. Section 3 defines the network and threat models. Section 4 develops the model in full. Section 5 describes the simulation campaign. Section 6 reports results, Section 7 interprets them, Section 8 states threats to validity, and Section 9 concludes.

2 Related Work

2.1 Trust-based detection and routing

Trust management in MANETs has developed along two largely separate lines. The first evaluates node reliability from forwarding behaviour, reputation and recommendation, and uses the resulting score to steer route selection [8–11]. Its foundations are the watchdog and pathrater of Marti et al. [12], which overhear whether a neighbour forwards what it receives, and CONFIDANT [13], which turns such observations into a Bayesian reputation and isolates nodes whose reputation falls below a threshold. Approaches in this line combine direct observation with neighbour recommendation [11, 14] and, more recently, with learned models [15, 16]. The recurring design questions are how to weight one's own observations against those of others, how to age evidence, and how to resist a recommender that lies.

Three design axes recur. The first is what the trust score is computed from. Early work used packet-forwarding ratio and drop behaviour [9, 14]. Later schemes added energy, mobility and link stability [17, 18], and recent ones learn the score from features using gradient boosting or neural models [5, 15]. The second is what the score is used for. Most schemes feed it into route selection, either by penalising low-trust hops in a single-path protocol [19, 20] or by distributing traffic across trust-weighted multiple paths [10, 21, 22]. A smaller group treats trust as an admission-control question, authenticating or excluding nodes before they participate [23, 24]. The third is how recommendations from other nodes are weighted and filtered, since a recommender may itself be adversarial [11, 25, 26].

Our work sits in the second and third of these but differs in what it optimises. A recent systematic review of malicious-node handling in wireless multi-hop networks found that more than 90% of the surveyed studies address detection and very few address removal [27]. Almost all of the above evaluate the trust mechanism by its effect on delivery ratio, throughput and delay, which measures the benefit of excluding attackers. Far fewer measure the cost of excluding honest nodes by mistake, and fewer still measure how long such an exclusion persists. Schemes that also address energy and route stability [18, 28] make the cost side explicit for energy but not for availability. We argue in Section 4 that a cumulative false-isolation measure belongs alongside the false-positive rate, and Section 6 shows the two can differ by more than an order of magnitude for the same detector.

A consistent finding across this line is that static thresholds produce false positives under changing topology, which motivates dynamic or adaptive trust state. Chatzidakis and Hadjiefthymiades [3] make this explicit, proposing a mechanism that identifies malicious nodes, alerts the network, and subsequently restores trust when a node resumes healthy behaviour or when an earlier classification proves incorrect. We adopt the possibility of restoration, and Section 6.11 shows that it is harder to use safely than it appears: an

isolated node cannot display the healthy behaviour that restoration waits for, so a rule that restores trust when bad evidence stops can re-admit an attacker.

2.2 Learned trust models

A recent line of work replaces hand-specified trust arithmetic with learned models. Gradient-boosted trees have been applied to trust-value estimation for malicious node detection [15], feature-optimisation and classification pipelines to intruder isolation [29], and reinforcement learning to reputation-driven route selection [5]. The attraction is that the decision boundary is fitted rather than assumed, which matters when the relationship between evidence and misbehaviour is not monotone or not separable by a threshold.

The coverage failure reported in Section 6 is not, however, a modelling failure. Under the false-location attack the delivered-throughput trace is identical to the clean-network trace in every seed, so the behavioural evidence stream contains no information about the attack. No estimator recovers signal from an identical trace, and a learned model restricted to that evidence class would fail exactly as the statistical one does. Evidence coverage is a property of what is collected, not of how it is combined. A learned rule could contribute at the combination stage; it cannot address a limitation upstream of it.

2.3 Localisation-centred beacon security

The second line is beacon-specific, using ranging measurements and geometric consistency to determine whether an advertised position is plausible [2, 30]. Robustness here usually means tolerating a bounded number of lying beacons rather than identifying them individually, and the classical results bound achievable localisation error as a function of the fraction of compromised anchors. Recent work continues this line for static sensor networks: BO-MAP combines clustering with probabilistic analysis to predict and isolate malicious anchors [31], and Tomic et al. detect spoofing anchors by voting over anchor-pair intersections, deliberately excluding no anchor because a wrongly removed honest anchor degrades accuracy [32]. Both assume static anchors and a single snapshot of ranging evidence, and neither uses forwarding behaviour or models a reversible decision.

Position falsification is also studied in vehicular networks, where misbehaviour detection checks the plausibility of the positions vehicles broadcast. The VeReMi dataset made such detectors comparable on a common set of falsification attacks [33], and recent detectors learn from message sequences, for example with federated Transformers [34]. The evidence class is the same as our localisation branch, but these systems detect misbehaving senders and leave revocation to a credential authority. They neither combine positional with forwarding evidence nor measure what a wrongful revocation costs.

Secure localisation has a longer history in sensor networks. SeRLoc bounds a node's position using directional beacons without range measurements [35]; verifiable multilater-

ation rejects position claims inconsistent with distance bounds [36]; and Liu, Ning and Du detect malicious beacons by the inconsistency between measured and declared range, revoking a beacon once a base station has received enough independent alerts [1]. Our localisation branch uses the same residual as the last of these. What differs is the setting — mobile beacons and commodity 802.11 RSSI rather than static beacons or UWB ranging — and the decision rule: our confirmation is reversible, and we measure what a false confirmation costs while it persists.

The two lines overlap less than their subject matter suggests, and the gap is visible in the evaluations rather than in the methods: a trust-routing paper does not typically test an attacker whose only misbehaviour is positional, and a beacon-security paper does not typically test one whose only misbehaviour is in forwarding. This is an observation about the papers we surveyed, not a systematic survey finding.

2.4 EMBN-MANET

EMBN-MANET [2] is the recent work closest to ours. Its detection evidence consists of RSSI measurements and trilateration in a UWB-based mobile ad hoc network. A k-polytopes procedure identifies centroids, from which an identifier infers suspicious activity, and a valid probable location T_{loc} is compared against the location F_{loc} a suspect beacon shares during coordinated localisation. Its threat model is a beacon that masquerades or spoofs by providing false location coordinates. Its response is to assign a reputation value to the identified beacon and to exclude it from coordinating localisation once that value crosses a threshold. Reported identification is complete when fewer than four malicious beacons occupy a neighbourhood and exceeds 85% above that count. The authors state that their work addresses static beacon nodes and name detection for mobile beacons as future work.

Four differences follow. First, EMBN-MANET's evidence is entirely localisation-derived. A beacon that advertises correct coordinates while selectively dropping transit traffic falls outside what its measurements can observe, and outside the threat model it sets out to address. Second, its response excludes a beacon from localisation coordination. It does not remove the node from packet forwarding, so it makes no claim about post-removal route continuity or throughput, and it is not a model of the forwarding-level isolation we evaluate. None of our isolation policies is an implementation of its response. Third, its beacons are static and ours are mobile, which its authors identify as the natural extension. Fourth, its accuracy figure is conditional on attacker density within a neighbourhood, and we quote it as such.

Table 1: Evidence classes and observable attack surface by approach family. Entries are read from published descriptions, not measured.

Approach	Loc. evid.	Beh. evid.	Rec. trust	Mob. beac.	FI metric
Localisation-centred [1, 2, 30]	✓				
Behavioural trust [9, 14]		✓	✓	✓	
Multipath trust [10, 21]		✓	✓	✓	
Learned trust [5, 15]		✓		✓	
Admission control [23, 24]		✓	✓	✓	
This work	✓	✓	✓	✓	✓

Loc./Beh. evid.: localisation and behavioural evidence. Rec. trust: recommendation-based trust. Mob. beac.: mobile beacons. FI metric: reports a false-isolation cost measure.

2.5 Structured comparison and basis for comparison

Alongside the measured baselines of Section 6.6, we give a structured qualitative comparison. Table 1 states, for each family of approach, which evidence classes it observes and which attack classes therefore fall inside its scope. Localisation-centred beacon security observes the evidence class that the false-location attack disturbs and not the class that selective forwarding disturbs; behavioural trust schemes observe the converse. Our contribution is not a better detector within either column but the observation that the columns are disjoint and that a beacon can misbehave in whichever one is not being watched. The final column is the metric introduced in Section 4; we are not aware of prior work in this area that reports the persistence of false exclusions rather than their rate, though we offer that as a gap in our reading, not an exhaustive search.

Reported detection accuracy in this field is conditioned on an attack model, a topology, a mobility regime, a traffic load and an attacker density, all of which vary across papers. It also saturates: when published figures cluster between 90% and 99%, a difference of one percentage point carries less information than the differences in evaluation conditions that produced it. Our own results show why. Every configuration tested, including both single-evidence ones, achieves perfect recall against the attack its evidence can observe and near zero against the attack it cannot. An accuracy figure drawn from the first case alone would be correct and uninformative. The informative axis is coverage. We compare against EMBN-MANET on it analytically, and Section 6.6 measures it for re-implemented decision rules of both families on identical traces.

3 Network and Threat Model

3.1 Network model

The network is a dynamic graph $G(t) = (V, E(t))$ over N mobile nodes with time-varying wireless links. A subset $\mathcal{B} \subseteq V$ are beacon nodes advertising position information usable

by neighbours for localisation. A subset $\mathcal{B}_m \subseteq \mathcal{B}$ are compromised. Each monitored beacon i carries a state

$$S_i(t) \in \{\textsc{Normal}, \textsc{Suspicious}, \textsc{Confirmed}\}. \tag{1}$$

The intermediate state exists so that a single anomalous observation does not directly authorise removal.

The pipeline both raises suspicion and decides what follows it; the contribution concerns the second step, the verification and response that follow a first suspicion. Attackers are insiders holding valid credentials, so cryptographic authentication does not resolve the problem: an authenticated node can still lie.

3.2 Attack model

We evaluate three attacker behaviours, chosen so the first two are disjoint in the evidence they disturb and the third combines them.

A1, false location. The compromised beacon advertises a position displaced from its true position, while forwarding all transit traffic honestly. This corrupts localisation and, by construction, produces no forwarding anomaly. The offset δ is applied to both coordinates and the result is clamped to the field,

$$x^c = \min\Big(X, \max(0, x + \delta)\Big), \qquad y^c = \min\Big(Y, \max(0, y + \delta)\Big). \tag{2}$$

An unclamped beacon is therefore displaced by $\delta\sqrt{2} = 424\,\text{m}$ for $\delta = 300\,\text{m}$, while one near the far corner is displaced much less. Over the campaign the realised displacement averaged 267 m, with median 271 m and 95th percentile 299 m (Section 6.7). We report the measured displacement alongside the nominal parameter, since the effective attack strength depends on where in the field the attacker happens to be.

A2, selective forwarding. The compromised beacon advertises its true position and drops transit packets with probability p. This degrades delivery and produces no localisation anomaly.

A3, hybrid at half intensity. The compromised beacon applies both behaviours with each halved: offset $\delta/2$ and drop probability $p/2$.

The choice to halve rather than compound is deliberate. Compounding both behaviours at full strength produces an attacker that is more conspicuous on every axis at once, which tests nothing A1 and A2 do not already test. Halving produces an attacker that is less conspicuous on each axis while remaining malicious on both. That is the configuration under which a fusion advantage, if the mechanism confers one, should appear, because each single-evidence branch operates closer to its detection threshold. A3 is therefore a test of the fusion hypothesis rather than a demonstration of it, and Section 6.5 sweeps its

Table 2: Principal notation.

Symbol	Meaning
$FR_i(t)$	forwarding ratio of beacon i
z_B, z_L	standardised behavioural / localisation evidence
A_B, A_L	oriented anomaly scores
$Z_{ij}(w)$	fused evidence statistic
T^D, T^I, T^H	direct, indirect, historical trust
$\hat{C}_i(t)$	collapsed confidence
τ_N, τ_M	normal and malicious thresholds
W, K	persistence window and confirmation count
$\mathcal{B}_{iso}(t)$	isolation set
θ_B, θ_L	branch detection thresholds in attack intensity
R_{TH}	throughput recovery ratio
$\Phi, \bar{\Phi}$	cumulative false isolation (node-seconds) and its normalised form

intensity.

4 Proposed Framework

This section develops the verification and recovery model in full. Fig. 1 shows how the pieces connect; notation is summarised in Table 2.

4.1 Behavioural evidence

For beacon i in window w, let P_i^{fwd} and P_i^{recv} denote packets forwarded and received for forwarding. The behavioural observable is the forwarding ratio

$$FR_i(t) = \frac{P_i^{fwd}(t)}{P_i^{recv}(t) + \epsilon}, \tag{3}$$

with ϵ guarding against division by zero.

We use a single behavioural observable rather than a composite score, because composites in this literature are often degenerate. A weighted sum of the form $w_1 FR_i + w_2(1 - DR_i) + \ldots$, where $DR_i = 1 - FR_i$ is the drop ratio, collapses to $(w_1 + w_2)FR_i$: what reads as two independent pieces of evidence is one observable counted twice, with an inflated effective weight. Observables genuinely independent of FR_i, such as routing-response consistency or response delay, would each require their own null calibration and are outside the scope of this study.

The observable is standardised against contemporaneous peers rather than a static calibrated baseline. Benign congestion depresses forwarding across all nodes in a neighbourhood simultaneously; measured against a fixed baseline this registers as individual misbehaviour, whereas measured against peers observed in the same window a network-

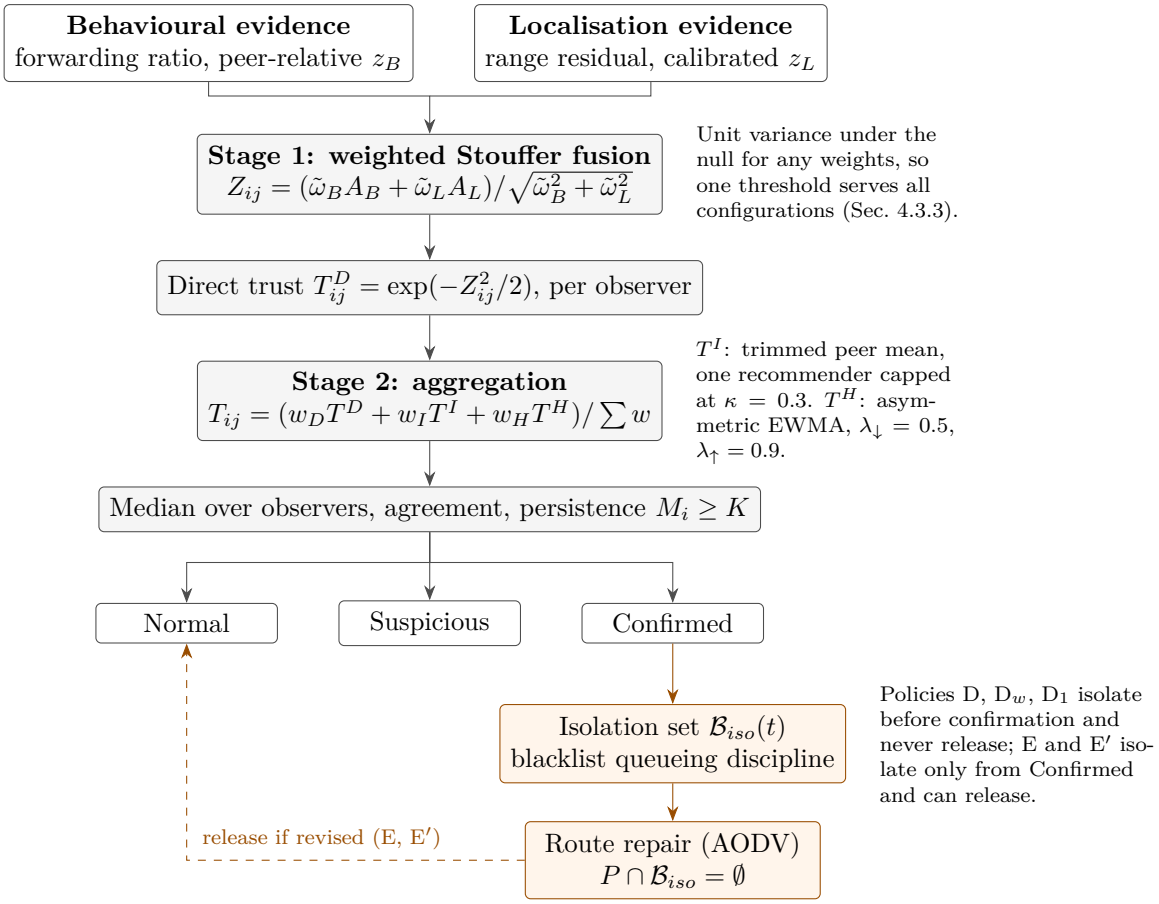


Figure 1: Verification pipeline. Behavioural and localisation evidence are standardised, oriented and fused in Stage 1 to give a per-observer direct trust value; Stage 2 aggregates that value across observers and over time. The three-state verdict separates suspicion from confirmation, and the isolation policies differ only in which evidence authorises removal and whether removal can be reversed.

wide depression cancels. If $\mathcal{P}_i$ is the peer set of i with at least n_{min} members,

$$z_B(w) = \frac{FR_i(w) - \mu(\mathcal{P}_i, w)}{\sigma(\mathcal{P}_i, w) + \epsilon}, \tag{4}$$

falling back to the static calibrated null where the peer set is too small to estimate. Equation (4) is the quantity that enters fusion.

4.2 Localisation evidence

4.2.1 Range residual

The detector is not given the beacon's true position: a node in a MANET has no such oracle, and a model that consumes one is not implementable. We therefore score the internal inconsistency of what the beacon broadcasts, which requires only quantities an observer can measure.

For observer j receiving a beacon from i, let $\hat{d}_{ij}$ be the distance implied by measured RSSI under the calibrated path-loss model, and $\tilde{d}_{ij}$ the distance implied by the coordinates the beacon advertises together with j's own position estimate. The range residual is

$$r_{ij}(w) = \left|\hat{d}_{ij}(w) - \tilde{d}_{ij}(w)\right|, \tag{5}$$

and the standardised localisation evidence is r_{ij} scored against the residual null distribution calibrated on clean-network runs,

$$z_L(w) = \frac{r_{ij}(w) - \mu_r}{\sigma_r + \epsilon}. \tag{6}$$

A beacon displaced by δ inflates $\tilde{d}_{ij}$ without changing $\hat{d}_{ij}$, so r_{ij} grows with δ; an honest beacon leaves r_{ij} at the measurement noise floor. True positions appear in our logs only for labelling detections as correct or incorrect after the fact.

Two consequences follow. First, the residual is bounded below by RSSI ranging error, so the attack is undetectable for offsets comparable to that error; the detection floor is a property of the ranging channel and not of the fusion rule. Second, an attacker who could also manipulate the RSSI observed by its neighbours could suppress the residual. Our threat model does not include physical-layer power manipulation, and we flag it as a gap rather than a solved case.

4.3 Stage one: evidence fusion

4.3.1 Orienting the two evidence streams

The two standardised statistics point in opposite directions with respect to misbehaviour. A malicious beacon depresses its forwarding ratio, so z_B becomes large and negative; it inflates its range residual, so z_L becomes large and positive. Summing the two as they stand would allow a node misbehaving on both axes to cancel its own evidence. We therefore define a common one-sided anomaly scale on which larger means more suspicious,

$$A_B(w) = -z_B(w), \qquad A_L(w) = +z_L(w), \tag{7}$$

and all subsequent fusion operates on A_B and A_L. The sign convention is trivial once stated, but Section 6.2 shows that getting it wrong produces a detector that fails specifically on hybrid attacks while continuing to work on single-axis ones.

4.3.2 Confidence-weighted fusion

Each branch is weighted by a fixed branch weight and by a confidence factor saturating in sample count,

$$c(n) = \frac{n}{n + n_{1/2}}, \tag{8}$$

with $n_{1/2} = 5$ for localisation and $n_{1/2} = 45$ for behaviour. A single beacon reception carries positional information directly, whereas forwarding behaviour is observable only through an accumulation of transit packets. This asymmetry reappears in the measured confirmation latencies of Section 6. Writing the effective weights as

$$\tilde{\omega}_B(w) = \omega_B \, c(n_B(w)), \qquad \tilde{\omega}_L(w) = \omega_L \, c(n_L(w)), \tag{9}$$

where n_B and n_L are the observation counts available to each branch in window w, the fused statistic is

$$Z_{ij}(w) = \frac{\tilde{\omega}_B A_B(w) + \tilde{\omega}_L A_L(w)}{\sqrt{\tilde{\omega}_B^2 + \tilde{\omega}_L^2}}. \tag{10}$$

A branch with no observations in a window has $c = 0$ and drops out, leaving the remaining branch unchanged; this is why the confidence factor appears inside the normalisation rather than outside it.

Direct trust is the fused statistic mapped through the Gaussian kernel

$$T_{ij}^D(w) = \phi\Big(Z_{ij}(w)\Big) = \exp\Big(-\tfrac{1}{2} Z_{ij}(w)^2\Big), \tag{11}$$

so that $T^D = 1$ for a beacon indistinguishable from the calibrated null and $T^D \to 0$ as the fused statistic moves into either tail. Thresholds on T^D therefore correspond to thresholds on $|Z|$: the values used here, $\tau_M = 0.40$ and $\tau_N = 0.70$, correspond to $|Z| = 1.354$ and $|Z| = 0.845$, two-sided null tail probabilities of 0.176 and 0.398.

Two properties of (11) matter. It is even in Z, so a deviation in either direction reduces trust equally; this is deliberate, since a beacon forwarding implausibly more traffic than its peers is as anomalous as one forwarding less. And because ϕ is applied after fusion rather than before it, the two-sidedness does not rescue a sign error: it is the fused Z that is folded, not the individual branches.

4.3.3 Calibrating the fused statistic

The combination rule matters less than how its output is thresholded. Let $s_B, s_L \in [0, 1]$ be evidence scores and $C = \omega_B s_B + \omega_L s_L$ with $\omega_B + \omega_L = 1$. Then

$$\min(s_B, s_L) \le C \le \max(s_B, s_L), \tag{12}$$

so if both scores lie on the benign side of a threshold τ, so does C. A detector that applies the single-branch threshold to a convex combination therefore cannot confirm an attacker that both branches miss. Tables built this way will still show the fused row ahead of the weaker branch, because a mean of a strong score and a weak one exceeds the weak one; the apparent advantage is a property of the comparison, not of the mechanism.

The bound is a consequence of reusing τ, not of averaging as such. For two independent standardised branches the arithmetic mean $\bar{z} = (z_B + z_L)/2$ has standard deviation $1/\sqrt{2}$ under the null, so a threshold recalibrated to the same false-positive rate is $\tau/\sqrt{2}$, and the rule $\bar{z} > \tau/\sqrt{2}$ is identical to the equal-weight Stouffer rule $(z_B + z_L)/\sqrt{2} > \tau$. What the fusion hypothesis requires is a combined statistic calibrated against its own null distribution. Weighted Stouffer combination (10) is unit variance under the null whatever the weights, so a single threshold serves every configuration and the hypothesis becomes falsifiable: two branches each at $z = 1.5$ combine, with equal weights, to $Z = 2.12$.

This argument disqualifies uncalibrated combination; it does not make Stouffer's method the best calibrated rule, and we do not claim that it is. Fisher's method, likelihood-ratio and Bayesian fusion, Dempster–Shafer combination [11], and learned rules can all be calibrated in the same sense. We chose Stouffer's method because it accepts continuous evidence without a likelihood model for either branch, admits per-branch weights directly [7], and degrades gracefully when one branch has no observations in a window, which is frequent under mobility. Section 6.5 compares it against a disjunctive rule; the remaining rules are listed in Section 9.1.

Equation (10) assumes z_B and z_L are independent under the null. In our attack scenarios they are close to independent by construction, because A1 and A2 disturb disjoint observables. Under a correlated failure, such as congestion coinciding with a mobility-induced ranging error, the fused statistic would be optimistic; correcting for it requires an estimated correlation term.

4.4 Stage two: trust aggregation

4.4.1 Direct trust

Direct trust is the Stage 1 output $T_{ij}^{D}(w)$ of (11), evaluated per observer per window. It is not separately smoothed; temporal weighting enters the model once, in T^{H} below.

4.4.2 Indirect trust

Let $R_{ij}(t) \in [0, 1]$ be neighbour j's recommendation regarding beacon i and $T_j(t)$ the reliability of that recommender:

$$T_i^{I}(t) = \frac{\sum_{j \in N_i} T_j(t) R_{ij}(t)}{\sum_{j \in N_i} T_j(t) + \epsilon}. \tag{13}$$

Equation (13) is the general reliability-weighted form. The evaluated implementation uses a specialisation in which R_{ij} is the direct trust observer j holds in target i, and the reliability weights are replaced by a robust aggregation rule: a trimmed mean over the direct trust of observers other than j, with j's own assessment excluded to avoid self-reinforcement, trim fraction 0.2, and any single recommender's influence capped at $\kappa = 0.3$. The cap bounds the effect of a colluding recommender, following the robustness principle of attack-resistant location estimation [30]: no minority of recommenders can move the aggregate arbitrarily.

4.4.3 Historical trust

T^H is an exponentially weighted average of prior combined trust for the (j, i) pair with asymmetric decay. The update is applied after the combination of (15) for window t has been formed, and the value it contributes to that combination is the one carried forward from the previous window:

$$T^H_{ij}(t) = \lambda\, T^H_{ij}(t-1) + (1-\lambda)\, T_{ij}(t), \qquad \lambda = \begin{cases} \lambda_\downarrow = 0.5, & T_{ij}(t) < T^H_{ij}(t-1), \\ \lambda_\uparrow = 0.9, & \text{otherwise}, \end{cases} \tag{14}$$

so trust degrades faster than it recovers: wrongly trusting an attacker costs more than briefly distrusting an honest node. On the first window for a pair, T^H is undefined, drops out of the renormalisation, and is seeded from the resulting combination.

4.4.4 Combined trust

$$T_{ij}(t) = \frac{w_D T^D_{ij}(t) + w_I T^I_{ij}(t) + w_H T^H_{ij}(t-1)}{w_D + w_I + w_H}, \tag{15}$$

renormalised over whichever terms are defined in the window. Because T^H seeds from the combined value, it cannot be evaluated in isolation: a configuration weighting history alone yields no defined trust at any window. This constrains the ablation design of Section 6.12.

4.5 Confidence, state and confirmation

Per-observer trust is collapsed to a single per-target confidence by the median over observers holding a defined opinion in that window,

$$C_i(t) = \text{median}\{\, T_{ij}(t) : j \in O_i(t) \,\}, \tag{16}$$

where $O_i(t)$ is that observer set and $|O_i(t)| \geq 3$ is required for any verdict. The median prevents a single observer with an extreme value from moving the collapsed confidence. The confidence may optionally be smoothed, $\hat{C}_i(t) = \gamma C_i(t) + (1-\gamma)\hat{C}_i(t-1)$, though

the evaluated configuration uses $\gamma = 1$, so temporal memory enters the decision through the persistence requirement rather than through smoothing. The verdict map is

$$S_i(t) = \begin{cases} \textsc{Normal}, & \hat{C}_i(t) \geq \tau_N, \\ \textsc{Suspicious}, & \tau_M < \hat{C}_i(t) < \tau_N, \\ \textsc{Confirmed}, & \hat{C}_i(t) \leq \tau_M, \end{cases} \tag{17}$$

with $0 \leq \tau_M < \tau_N \leq 1$. Confirmation additionally requires persistence,

$$M_i(t) = \sum_{k=0}^{W-1} \mathbb{I}\Big[\hat{C}_i(t-k) \leq \tau_M\Big], \tag{18}$$

with $W = K = 3$, so beacon i is confirmed only if the three most recent windows all satisfy the condition. A verdict also requires agreement: at least a fraction $a_{min} = 0.5$ of observers holding a defined opinion, and no fewer than three observers, must place the target below τ_M in the same window.

Two further rules complete the decision logic.

Single-branch override. An observer whose direct trust falls below $\tau_X = 0.10$, corresponding to $|Z| \geq 2.146$, can confirm a target without satisfying the agreement fraction. An attack visible to only one evidence class produces one branch far into its tail while the other stays at the null, and requiring majority agreement there would suppress the detection the architecture is meant to enable. The override is load-bearing for the coverage result: under A1 the behavioural branch contributes nothing, and confirmation proceeds on localisation evidence alone. The A1 and A2 results should be attributed to fusion together with this override, not to fusion by itself.

Asymmetric clearing. Leaving the confirmed state requires $K_c = 12$ consecutive clean windows against the $K = 3$ required to enter it, a four-fold asymmetry in the same direction as the trust decay constants. This rule gives policy E its release behaviour.

Evidence-aware release. Isolation removes a beacon from forwarding, so while it is isolated its behavioural branch receives too few transit packets to contribute, and its trust rests on localisation alone. Under policy E a beacon confirmed on behavioural evidence can therefore be cleared by the absence of that evidence rather than by its return. Policy E$'$ closes this path: a beacon confirmed in a window in which behavioural evidence was available is released only on a clean window in which behavioural evidence is again available. A beacon confirmed on localisation evidence alone is released as under E. Section 6.11 measures what the rule costs.

4.6 Isolation

On confirmation the beacon enters the isolation set, and under a releasing policy it leaves when released,

$$\mathcal{B}_{iso}(t) = \Big(\mathcal{B}_{iso}(t-1) \cup \{\, i : S_i(t) = \textsc{Confirmed} \,\}\Big) \setminus \mathcal{R}(t), \tag{19}$$

where $\mathcal{R}(t)$ is the set released at t (empty for the policies that never release). The admissible node set becomes $V'(t) = V \setminus \mathcal{B}_{iso}(t)$. Every subsequent route must satisfy $P \cap \mathcal{B}_{iso}(t) = \emptyset$. Enforcement is by a blacklist queueing discipline; route repair is performed by AODV.

4.7 Evaluation quantities

Recovery time is $T_{rec} = T_{new\text{-}route} - T_{isolation}$. Standard network measures are

$$PDR = \frac{P_{received}}{P_{sent}}, \qquad TH = \frac{8D_{received}}{T_{sim}}, \qquad RO = \frac{N_{control}}{N_{data}}, \tag{20}$$

and we report the throughput recovery ratio

$$R_{TH} = \frac{TH_{after}}{TH_{normal}}, \tag{21}$$

for which a well-behaved mechanism gives $R \to 1$.

Finally we define a false-isolation cost that we believe is under-reported in this literature. Let H be the set of honest beacons and $\iota_i(t)$ the indicator that i is isolated at time t. Cumulative false isolation over a run of duration T is

$$\Phi = \int_0^T \sum_{i \in H} \iota_i(t)\, dt, \tag{22}$$

in node-seconds. Φ measures service withdrawn from the network by the security mechanism itself. A mechanism with a low false-positive rate may still have large Φ if it never releases. Because Φ grows with the number of honest beacons and with run duration, we also report the dimensionless form

$$\bar{\Phi} = \frac{\Phi}{|H| \cdot T}, \tag{23}$$

the mean fraction of honest-beacon service time withdrawn by the mechanism, which is comparable across topologies.

Table 3: Network and simulation parameters.

Parameter	Symbol	Value
Simulator	—	ns-3.48
Routing protocol	—	AODV
Field	—	500×500 m (sparse: 1000×1000; 50-node: 707×707)
Nodes	N	25 (50-node campaign: 50)
Monitored beacons	$\lvert\mathcal{B}\rvert$	3, 1 attacker (50-node campaign: 6, 2 attackers)
Transmit power	—	30 dBm
Propagation loss model	—	log-distance
Path-loss exponent	η	3.0
Reference loss at 1 m	P_0	46.68 dB
Shadowing	σ_{sh}	0 dB; 4 and 8 dB in Sec. 6.13
RSSI measurement noise	σ_{RSSI}	4 dB
Mobility model	—	random waypoint
Node speed	—	$U(1, 5)$ m/s
Pause time	—	2.0 s
Traffic	—	10 CBR flows, 12 pkt/s (50-node campaign: 14)
Packet size	—	512 B
Beacon interval	—	1.0 s
PHY data rate	—	DSSS 1 Mbps
Simulation duration	T	400 s
Attack start	—	$t = 30$ s
Per-axis position offset (A1)	δ	300 m
Measured mean displacement (A1)	—	267 m
Drop probability (A2)	p	0.6
Hybrid scaling (A3)	—	$\delta/2$, $p/2$
Runs executed	—	780 (Table 5)

5 Experimental Methodology

Simulations use ns-3.48 [37] with AODV routing over IEEE 802.11. Network parameters appear in Table 3 and detector parameters in Table 4; both were fixed for the entire campaign and no per-scenario tuning was performed.

Two properties of the radio configuration bound what the localisation branch is asked to do. First, the primary campaign disables shadowing, so the channel is deterministic log-distance path loss and the only stochasticity entering the localisation branch is the 4 dB measurement noise added at the receiver. A real deployment would add multipath, obstruction and correlated shadowing, all of which inflate the range residual for honest beacons and raise the detection floor; Section 6.13 repeats the campaign at $\sigma_{sh} = 4$ and 8 dB. Second, the observer inverts distance from measured RSSI using the same path-loss parameters,

$$\hat{d}_{ij} = 10^{(P_{tx} - P_0 - \mathrm{RSSI}_{ij})/(10\eta)}, \tag{24}$$

Table 4: Detector parameters. All values fixed across the campaign; none were tuned per scenario.

Parameter	Symbol	Value
Observation window / step	—	10 s / 5 s
Branch weights	ω_B, ω_L	1.0, 1.0
Confidence half-weight, localisation	$n^L_{1/2}$	5
Confidence half-weight, behaviour	$n^B_{1/2}$	45
Minimum peer set	n_{min}	8
Minimum transit packets, behaviour	—	5
Trust weights	w_D, w_I, w_H	0.5, 0.2, 0.3
Trim fraction / recommender cap	— / κ	0.2 / 0.3
History decay, falling / rising	$\lambda_\downarrow$ / $\lambda_\uparrow$	0.5 / 0.9
Observer agreement	a_{min}	0.5
Malicious threshold	τ_M	0.40 ($\lvert Z\rvert = 1.354$)
Normal threshold	τ_N	0.70 ($\lvert Z\rvert = 0.845$)
Persistence / clearing persistence	K / K_c	3 / 12 windows
Minimum observers	—	3
Single-branch override	τ_X	0.10 ($\lvert Z\rvert = 2.146$)
Bounded transform	ϕ	$\exp(-Z^2/2)$, Eq. (11)

so $\hat{d}_{ij}$ in (5) is derived entirely from a quantity the observer measures. No true position enters the detector.

The full-intensity attack magnitudes are large. A position offset of $\delta = 300$ m in a 500×500 m field is a very large lie, and a drop probability of $p = 0.6$ is an aggressive grey hole. Both were chosen to establish the mechanism rather than to represent a realistic adversary; Section 6.5 sweeps the hybrid attack down to a displacement of 3.75 m and a drop probability of 0.75%.

The sparse campaign keeps every parameter of Table 3 but quadruples the field area, so that nodes have about a quarter as many neighbours and routes are longer and break more often; detector thresholds are recalibrated on its own clean seeds. The 50-node campaign keeps node density constant: twice the nodes in twice the area, six monitored beacons of which two attack, and 14 traffic flows. Its thresholds are likewise recalibrated on its own clean seeds.

Table 5 lists the campaign. Three groups of runs repeat another part of it: the control arm of the enforced runs (15 runs) repeats the original detection runs, those detection runs (20) are seeds 1–5 of the 30-seed campaign, and the primary sweep's full-intensity level (10) is seeds 1–10 of the 30-seed A3 scenario. Every repeated run reproduced its counterpart's output files exactly, which we report as a determinism check, and a run repeated on a second machine with a separately built simulator reproduced the original output byte for byte; no analysis uses the same simulation twice. Throughput recovery, mechanism accounting, localisation error and the aggregation ablation come from the original campaign, in which isolation is enforced by the simulator. Coverage, detection

Table 5: Simulation campaign. Calibration seeds are clean runs held out from testing. Of the first 570 runs, 45 repeat a configuration and seed run elsewhere in the campaign and reproduced it byte for byte; closed-loop rounds that re-enforce an unchanged schedule likewise repeat their predecessor.

Part	Design	Runs
Original campaign	4 scenarios × 5 seeds (detection) and 3 attacks × 3 policies × 5 seeds with isolation enforced in the simulator	65
30-seed campaign	4 scenarios × seeds 1–30, plus calibration seeds 101–105	125
Intensity sweep, primary	A3, 11 levels × 10 seeds, drop $= 0.3f$	110
Intensity sweep, second pairing	A3, 8 levels × 10 seeds, drop $= 0.15f$	80
Shadowing	$\sigma_{sh} \in \{4, 8\}$ dB, 4 scenarios × 20 seeds, plus 5 calibration seeds each	170
Clean enforced replay	policies D_1, E on clean seeds 1–10, schedules from the replay enforced in the simulator	20
Iterated closed loop	policies E, E′ × {A2, clean} × seeds 1–10 × 3 rounds	120
Sparse topology	25 nodes in 1000 × 1000 m, 4 scenarios × 10 seeds, plus 5 calibration seeds	45
50-node network	50 nodes in 707 × 707 m, 6 monitored beacons, 2 attackers, 4 scenarios × 10 seeds, plus 5 calibration seeds	45
Total		780

latency, false positives and cumulative false isolation come from the 30-seed campaign. For the latter, each isolation policy is applied to the same per-window verdict stream offline: Φ depends only on when each beacon is isolated and released, so one detection run per seed yields every policy's Φ with the seeds paired across policies. The replay assumes that an isolation does not alter the verdicts issued after it. How far that assumption reaches differs by policy. D, D_w and D_1 never release, so the contribution of a beacon to Φ is fixed by its first isolation time, and that time is computed exactly from verdicts issued before any isolation occurred; feedback can shift only the isolation times of beacons removed later in the same run. Policy E releases on a revised verdict, so its release times, and hence its Φ, depend on verdicts issued while another beacon may be isolated. Removing a beacon from forwarding shifts load onto its neighbours and can move their peer-relative scores in either direction. The enforced runs of the original campaign are all attacked runs and do not test the clean-network replay, so we tested it separately: for clean seeds 1–10 the replayed D_1 and E schedules were enforced in the simulator, the enforced traces were re-verified with the same calibration, and each policy's schedule was rebuilt from the verdicts produced under isolation (20 further runs). This is one step of a fixed-point

check rather than a fully online detector, and Section 6.10 reports the result. For E and E′ we then iterated the loop: each round enforces the schedule rebuilt from the previous round's enforced verdicts, for three rounds on the A2 and clean scenarios (Section 6.11). We compare six isolation policies:

- **C, no isolation.** No node removed. Control.
- **D, immediate isolation.** Isolate on the first window in which a beacon is Suspicious or Confirmed; never release.
- **D_w, immediate with warm-up guard.** As D, but a beacon's first three usable windows cannot trigger isolation. The guard is counted per beacon from its own first usable window, never from the attack start.
- **D_1, single window of malicious evidence.** Isolate on the first window in which confidence falls below τ_M with enough observers in agreement, without the persistence requirement; never release. This is the strongest honest immediate policy.
- **E, confirmation-gated isolation.** Isolate only on confirmation; release when the verdict is subsequently revised.
- **E′, confirmation-gated isolation with evidence-aware release.** As E, but a beacon confirmed while behavioural evidence was available is released only when that evidence returns (Section 4). E′ was introduced after the enforced runs showed that E re-admits forwarding attackers (Section 6.11); because a beacon confirmed on behaviour stays isolated under E′, its closed-loop schedule follows exactly from the verdicts issued before isolation.

Policies D, D_w and D_1 are constructions for comparison and are not models of any published method; in particular they are not EMBN-MANET, whose response operates on localisation coordination rather than packet forwarding. Isolation is enforced by a blacklist queueing discipline installed on every node, including in control runs. With an empty blacklist the discipline behaves as a plain FIFO, so its queueing behaviour is present in the baseline as well.

5.1 Literature baselines

Two baselines re-implement the evidence class and decision rule of a well-known design and run on the same traces as our pipeline. They are not reimplementations of complete published systems.

LND, localisation-only alert counting. After Liu, Ning and Du [1], the evidence class EMBN-MANET also uses. In each window an observer with at least three receptions

raises an alert against a beacon whose mean range residual exceeds ρ, and the beacon is revoked permanently once more than τ' alerts have accumulated.

WD, behaviour-only reputation. After the watchdog of Marti et al. [12] and the Beta reputation of CONFIDANT [13]. A beacon's reputation is the mean of a Beta distribution whose parameters are the counts of forwarded and unforwarded transit packets, faded by 0.9 per 10 s window; once at least 20 packets of evidence have accumulated, the beacon is isolated permanently when its reputation falls below τ_W.

LND+WD. Both detectors run side by side and a beacon is isolated when either fires. This is the disjunctive rule of Section 6.5 applied to two independently designed detectors.

All three thresholds come from the held-out clean calibration seeds of each channel and from nothing else: ρ is the 99th percentile of clean residuals, τ_W the 1st percentile of clean reputations, and τ' the smallest alert count at which no calibration run revokes any beacon. At 0, 4 and 8 dB this gives $\rho = 63$, 87 and 170 m, $\tau_W = 0.81$, 0.57 and 0.38, and $\tau' = 10$, 11 and 15. A sweep of τ' on the test runs, reported for transparency, shows the trade-off the calibration resolves: at $\tau' = 5$ LND revoked 31 of 90 honest node-seeds on the clean network, at 10 three, and at 20 none, while still revoking the A1 attacker in every seed.

5.2 Reproducibility

The simulator is ns-3.48 built with GCC on Ubuntu under WSL2, with the analysis pipeline in Python 3. Simulator sources, the analysis pipeline and the scripts for every campaign are public at `https://github.com/kraghu356/manet-beacon-trust`. Every run uses an explicit integer seed, so any run is reproducible from its scenario, policy and seed. Detection thresholds were fixed on the trust scale before any attack scenario was run, and the null distributions of A_B and A_L were calibrated on clean-network runs that are never used for testing: seeds 101–105 for the 30-seed and shadowing campaigns, recalibrated per channel. No threshold was chosen by reference to a result reported in this paper. Several simulator parameters are supplied at invocation rather than compiled in, so the released command lines, not the source defaults, define the experiment. Statistical intervals are percentile bootstrap intervals over seeds (10,000 resamples), paired across policies where policies share seeds.

6 Results

6.1 Evidence coverage

Table 6 and Fig. 2 report attacker confirmation by evidence configuration. The inversion between the two single-evidence rows admits a structural explanation. Under A1 the

Table 6: Attacker confirmation by evidence configuration, 30 seeds per cell.

Configuration	A1	A2	A3
Behavioural only	3/30	30/30	30/30
Localisation only	30/30	0/30	30/30
Fused	30/30	30/30	30/30

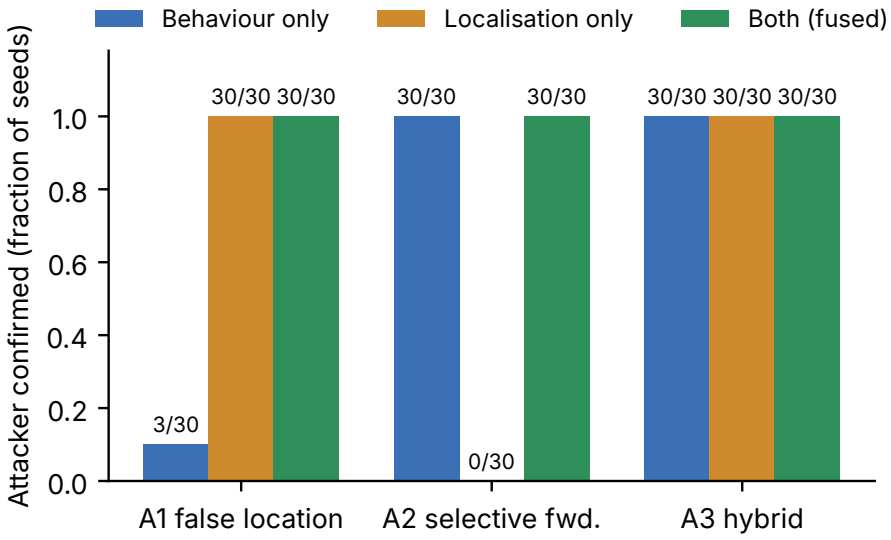


Figure 2: Evidence coverage over 30 seeds. Each single-evidence branch is blind to the attack that produces no signature in its class; the three behavioural confirmations of the A1 attacker occur at a lower rate than the same branch confirms honest beacons.

attacker forwards honestly, so no behavioural quantity departs from its null distribution. The delivered-throughput series of every A1 run is identical to the clean-network run of the same seed, in all 30 seeds (Fig. 3). Behavioural evidence alone nevertheless confirmed the A1 attacker in 3 of 30 seeds. These confirmations carry no attack signal: on the clean network the behavioural branch alone confirms honest beacons in 15 of 90 node-seeds (17%), a higher rate than its 10% on the A1 attacker. Symmetrically, under A2 the attacker advertises its true position, and localisation evidence alone confirmed it in 0 of 30 seeds; it confirmed no honest beacon in any run of the campaign.

A single-evidence detector therefore has a region of the attack space it cannot observe, and an adversary who knows which evidence is collected may choose to occupy it. This follows from what each evidence class can physically observe rather than from any threshold setting.

The scope of the claim is narrow. Table 6 establishes that collecting both classes covers both attacks. It does not establish that Stouffer fusion is responsible: a disjunctive rule that thresholds each branch independently and confirms on either would achieve the same result on A1 and A2, because in each case one branch is far into its tail while the other is silent. Fusion and disjunction separate only where both branches are weak simultaneously.

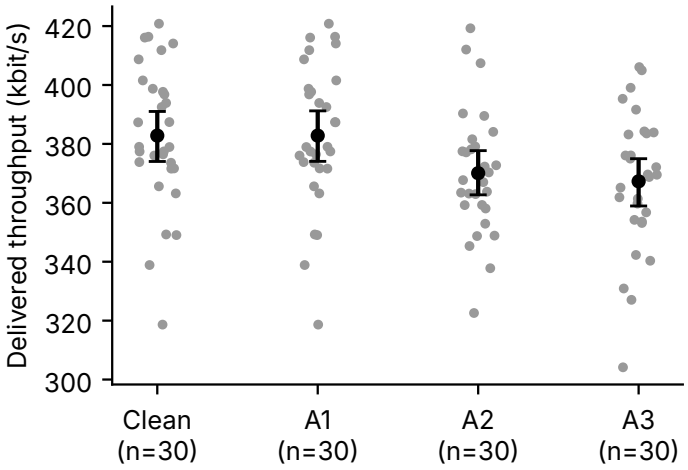


Figure 3: Delivered throughput by scenario, 30 seeds, with bootstrap 95% intervals of the mean. A1 and the clean network produce identical throughput in every seed: the false-location attack has no packet-level footprint.

The coverage result is a result about evidence collection; the fusion-rule question is taken up in Section 6.5.

Two timing quantities are reported and must not be conflated. Detection latency is the interval between attack onset and the first window in which the attacker is confirmed. Isolation time is the timestamp of the first isolate event written to the enforcement schedule. Over 30 seeds, the fused pipeline confirmed the attacker in every seed of every attack. Median latency was 15.0 s for A1 and A3 and 27.5 s for A2 (means 20.0, 20.7 and 41.7 s), with a tail of slower confirmations reaching 80, 75 and 160 s respectively.

The difference between attack classes is a property of the evidence, and the model predicts its direction. Behavioural evidence reaches half confidence weight at $n_{1/2} = 45$ observations against 5 for localisation (8). A2 is observable only through the behavioural branch, so its confidence factor climbs an order of magnitude more slowly and confirmation waits on the accumulation of transit packets. A1 and A3 are observable through localisation, where a handful of beacon receptions suffice. The constants were fixed during calibration and not adjusted afterwards.

6.2 Sensitivity to evidence orientation

The sign convention of (7) deserves its own analysis, because getting it wrong produces a detector that passes the tests most papers run and fails the one they do not. The transform (11) is even, but fusion happens before ϕ, so it is the summed statistic that is folded, not the individual branches.

Suppose the two statistics were summed without orientation, for a beacon whose behavioural anomaly reaches $|z_B| = 3$ and whose localisation anomaly reaches $z_L = 3$ at full intensity, with equal weights. Under A1, $z_B \approx 0$ and the misoriented fusion gives

$Z = +2.12$, hence $T^D = 0.10$: confirmed. Under A2, $z_L \approx 0$ and $Z = -2.12$, which because ϕ is even also gives $T^D = 0.10$: confirmed. Both single-axis attacks are caught. Under a hybrid attack at half intensity, however, $z_B \approx -1.5$ and $z_L \approx +1.5$, and

$$Z = \frac{(-1.5) + (+1.5)}{\sqrt{2}} = 0, \qquad T^D = \phi(0) = 1.0. \tag{25}$$

The attacker's two misbehaviours cancel exactly, and a node misbehaving on both axes receives perfect trust. A detector with this defect is blind to hybrid attacks in proportion to how evenly the attacker balances the two axes, and an adversary aware of the defect could balance them deliberately. With (7) applied, the same attacker gives $Z = \Big((+1.5) + (+1.5)\Big)/\sqrt{2} = 2.12$, recovering the full-intensity single-axis statistic from two half-intensity observations.

6.3 Attack intensity and the fusion region

All three configurations confirm the A3 attacker in every seed, so the experiment designed to show a fusion sensitivity advantage did not show one. To locate where an advantage could appear we assume that branch anomaly magnitude scales approximately linearly with attack intensity, $A(f) \approx fA(1)$. For an RSSI-derived range residual under mobility, and for a drop-rate statistic estimated from a finite window, that linearity is plausible over a limited range but not guaranteed, so what follows locates the region under an idealised model of the detector.

Let θ_B and θ_L be the intensities at which each branch alone reaches the decision threshold, so that branch X has anomaly $\theta f/\theta_X$ at intensity f. With effective weights $\tilde{\omega}_B, \tilde{\omega}_L$ the fused statistic of (10) crosses the threshold at

$$f_F = \frac{\sqrt{\tilde{\omega}_B^2 + \tilde{\omega}_L^2}}{\tilde{\omega}_B/\theta_B + \tilde{\omega}_L/\theta_L}, \tag{26}$$

and the family passes through a fusion-only region — an intensity at which fusion confirms and neither branch does — precisely when

$$f_F < \min(\theta_B, \theta_L). \tag{27}$$

When the branches are equally sensitive the region is $\theta_B/\sqrt{2} < f < \theta_B$: a factor of $\sqrt{2}$ in intensity, half an octave, and the widest it can be. Writing $r = \max(\theta_B, \theta_L)/\min(\theta_B, \theta_L)$ for the sensitivity ratio, (27) with equal effective weights reduces to

$$\frac{\sqrt{2}\, r}{1 + r} < 1 \quad \Longleftrightarrow \quad r < 1 + \sqrt{2} \approx 2.414. \tag{28}$$

With unequal weights, let $\tilde{\omega}_s$ weight the more sensitive branch and $\tilde{\omega}_w$ the less sensitive one; the region exists if and only if $r < \tilde{\omega}_w / \left(\sqrt{\tilde{\omega}_s^2 + \tilde{\omega}_w^2} - \tilde{\omega}_s\right)$, which falls below $1 + \sqrt{2}$ when the less sensitive branch carries more weight (1.618 at twice the weight).

The condition concerns the attack family, not the detector, and the distinction matters for how it is used. Write each branch's anomaly in units of the decision threshold, $u = A_L/\theta$ and $v = A_B/\theta$, so that a branch confirms alone when its coordinate reaches 1. With equal effective weights the fused statistic confirms when $u + v \geq \sqrt{2}$, so the fusion-only region is the triangle

$$u < 1, \qquad v < 1, \qquad u + v > \sqrt{2}, \tag{29}$$

which is non-empty for every equal-weight Stouffer detector, and with unequal positive weights the corresponding region is non-empty as well. Every such detector can therefore confirm attackers that neither branch confirms alone: those whose two anomalies are each sub-threshold and of comparable normalised size. Scaling an attack by intensity f moves it along the ray $(u, v) = f\,(1/\theta_L, 1/\theta_B)$ from the origin, and (28) is the condition for that ray to pass through the triangle. A lopsided attack, one that is much more visible to one branch than to the other, crosses the more sensitive branch's threshold while the other branch's anomaly is still too small to lift the fused statistic, and its ray misses the region. The ratio r is therefore a property of how the attack mixes the two misbehaviours, and it can be read from single-branch detection curves for a given attack family before any fusion experiment is run. Section 6.5 tests it on two families that differ only in that mix.

6.4 False positives

Over the 30-seed campaign the fused pipeline confirmed 9 of 90 honest node-seeds on the clean network (10%, Wilson 95% interval 5.4–17.9%) and 21 of 180 under attack (11.7%, 7.8–17.2%). The behavioural branch alone confirmed 15 of 90 on the clean network and 35 of 180 under attack; the localisation branch alone confirmed none. False confirmations are therefore a property of the behavioural branch, whose peer-relative score fires whenever a beacon's forwarding departs from its contemporaneous neighbours for benign reasons such as mobility, and fusion with a silent localisation branch damps rather than removes them. With three monitored beacons per run, one false positive moves per-run accuracy by 33 percentage points, which is why we report counts rather than accuracy.

6.5 Attack-intensity sweeps

We tested the fusion condition directly with two sweeps of the hybrid attack. The primary sweep scales displacement and drop probability together from the full-intensity A3 configuration over eleven levels, ten seeds each (110 runs, Table 7). The second repeats

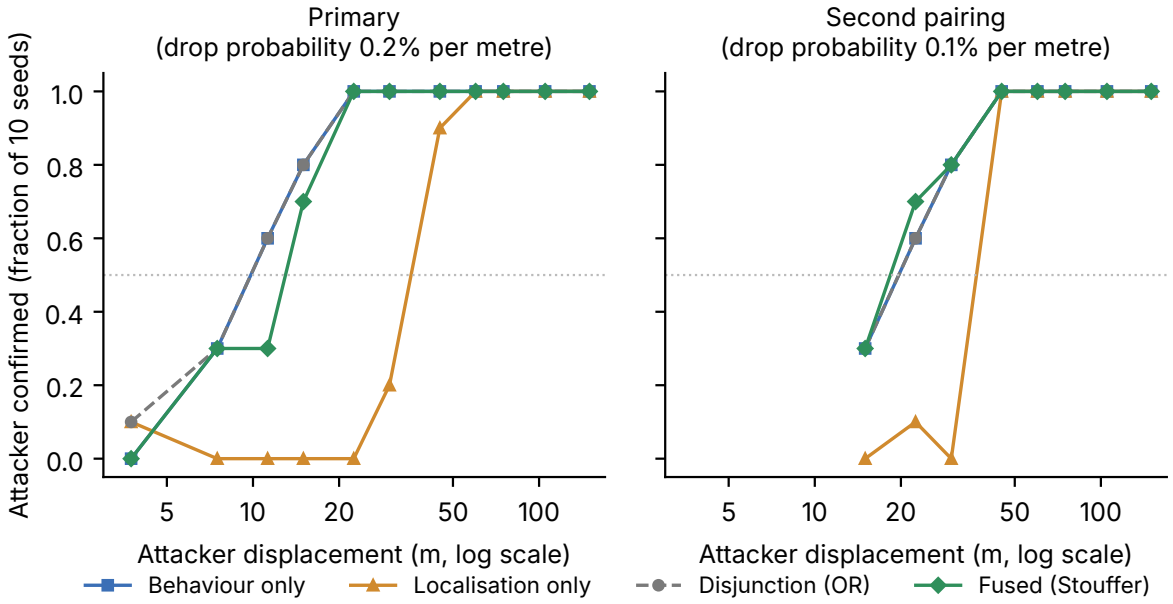


Figure 4: Attacker detection across the two intensity sweeps, ten seeds per point. Left: the branches' 50% thresholds lie 3.63 apart, beyond the $1 + \sqrt{2}$ bound, and fusion falls below the disjunctive rule near the behavioural threshold. Right: halving the drop rate moves the behavioural curve and brings the ratio to 1.87, inside the bound, and fusion matches or exceeds disjunction.

Table 7: Primary sweep: attacker confirmed (of 10 seeds). Displacement is the measured mean over the attacker's beacons after onset; below 45 m it equals the nominal $f\delta/2$ because the field boundary never clamps the offset. Drop probability is $0.3f$. OR is the disjunctive rule over the two single-evidence branches. FP counts honest beacons confirmed, of 20 node-seeds.

f	Displ. (m)	Drop	Beh.	Loc.	OR	Fused	FP (OR / fused)
0.025	3.75	0.0075	0	1	1	0	3 / 3
0.05	7.5	0.015	3	0	3	3	5 / 4
0.075	11.25	0.0225	6	0	6	**3**	1 / 1
0.10	15.0	0.03	8	0	8	**7**	5 / 4
0.15	22.5	0.045	10	0	10	10	5 / 5
0.20	30.0	0.06	10	2	10	10	4 / 1
0.30	45.0	0.09	10	9	10	10	4 / 4
0.40	60.0	0.12	10	10	10	10	2 / 2
0.50	74.9	0.15	10	10	10	10	3 / 2
0.70	104.5	0.21	10	10	10	10	3 / 1
1.00	147.1	0.30	10	10	10	10	3 / 3

eight levels with the drop probability halved relative to displacement (80 runs, Table 8). Fig. 4 shows both.

Table 8: Second pairing: attacker confirmed (of 10 seeds) with drop probability $0.15f$, half the rate of Table 7, at the same displacement.

f	Displ. (m)	Drop	Beh.	Loc.	OR	Fused	FP (OR / fused)
0.10	15.0	0.015	3	0	3	3	5 / 4
0.15	22.5	0.0225	6	1	6	**7**	1 / 1
0.20	30.0	0.03	8	0	8	8	5 / 4
0.30	45.0	0.045	10	10	10	10	5 / 5
0.40	60.0	0.06	10	10	10	10	4 / 1
0.50	74.9	0.075	10	10	10	10	2 / 2
0.70	104.4	0.105	10	10	10	10	3 / 2
1.00	147.1	0.15	10	10	10	10	3 / 2

6.5.1 Primary sweep: the attack path misses the fusion-only region

Behavioural detection falls from 8 of 10 at 15 m to 6, 3 and 0 of 10 at 11.25, 7.5 and 3.75 m; it reaches half detection at $\theta_B = 9.8$ m, interpolated on log intensity. Localisation evidence reaches half detection at $\theta_L = 35.7$ m. The measured ratio is $r = 3.63$. Resampling each level's ten seeds as independent Bernoulli trials gives a bootstrap 95% interval of [2.67, 5.47], and fewer than 0.5% of resamples fall below $1 + \sqrt{2}$. Since the less sensitive branch, localisation, carries the larger effective weight through (8), the applicable bound is tighter still. This attack family's path does not enter the fusion-only region.

The sweep shows the consequence the bound predicts. Where the behavioural branch is near its threshold, adding the weaker localisation branch dilutes it: at 11.25 m fusion confirmed the attacker in 3 of 10 seeds against 6 for the disjunctive rule, and at 15 m in 7 against 8; at every stronger intensity both confirmed it in all ten. The mechanism is visible at seed level. Where the behavioural branch is anomalous and the localisation branch sits at its null, equal-weight Stouffer combination computes $(A_B + 0)/\sqrt{2}$, attenuating a live branch by 29%, because a branch with adequate sample count carries confidence weight whether or not it carries signal. Disjunction is immune, since it never mixes the two.

6.5.2 Second pairing: the condition predicts the outcome

The ratio depends on how the sweep couples the two intensities, and the second sweep tests whether (28) predicts the consequence. The behavioural curve moved exactly as drop probability dictates: 3, 6 and 8 of 10 at drops of 1.5, 2.25 and 3% in both sweeps, so behavioural detection depends on the drop rate alone. Localisation, which depends on displacement alone, stayed at $\theta_L = 36.7$ m. With $\theta_B = 19.7$ m the ratio falls to $r = 1.87$ (bootstrap 95% interval [1.41, 2.21]), inside the equal-weight bound in every resample.

The outcome changed as the condition predicts. Fusion never confirmed fewer attackers than the disjunctive rule, and at $f = 0.15$, where the behavioural branch alone confirmed 6 of 10, fusion confirmed 7: one seed in which neither branch crossed its threshold but

Table 9: Pipeline against literature baselines on identical traces. Detection counts the attacker isolated after onset; FI is honest beacons isolated on the clean network; $\bar{\Phi}$ is clean-network false isolation (E′ for the pipeline); latency is the median time from onset to isolation, in seconds, for A1 / A2 / A3, given only where the attacker was detected in at least half the seeds. 30 seeds at 0 dB and 20 under shadowing.

σ_{sh}	Method	A1	A2	A3	FI	$\bar{\Phi}$	Latency A1 / A2 / A3
0 dB	LND	30/30	2/30	30/30	3/90	0.003	10 / — / 15
	WD	1/30	30/30	30/30	4/90	0.031	— / 20 / 40
	LND+WD	30/30	30/30	30/30	7/90	0.034	10 / 20 / 15
	Pipeline	30/30	30/30	30/30	9/90	0.050	15 / 27.5 / 15
4 dB	LND	20/20	1/20	20/20	2/60	0.005	15 / — / 22.5
	WD	4/20	20/20	13/20	7/60	0.050	— / 60 / 270
	LND+WD	20/20	20/20	20/20	8/60	0.054	15 / 60 / 22.5
	Pipeline	20/20	20/20	20/20	6/60	0.045	15 / 125 / 20
8 dB	LND	20/20	0/20	20/20	1/60	0.000	20 / — / 65
	WD	0/20	20/20	3/20	0/60	0.000	— / 120 / —
	LND+WD	20/20	20/20	20/20	1/60	0.000	20 / 120 / 65
	Pipeline	20/20	20/20	19/20	0/60	0.000	22.5 / 150 / 155

their combined evidence did. One seed is weak evidence and we do not report it as a fusion benefit. What the two sweeps establish is that the same detector dilutes or helps depending on how the attack family mixes its two misbehaviours, and that the ratio predicting which can be measured before fusion is applied. The effective weights vary by window through (8); if localisation carries twice the weight of behaviour, the bound tightens to 1.618, which 17% of resamples satisfy.

Taken together with Table 6, the finding is that evidence collection determines coverage, and whether the combination rule adds anything against a given attack family is decided by the branches' relative sensitivity to that family.

6.6 Comparison with literature baselines

Table 9 compares the pipeline with the baselines of Section 5.1 on identical traces. The pipeline's false isolation is that of policy E′, since the baselines also isolate on their own detection and, like E′ in practice, do not release.

The single-evidence baselines reproduce the coverage result with detectors we did not design. The localisation-only detector revoked the A1 and A3 attackers in every seed of every channel and the A2 attacker in at most 2 of 30, and those few revocations came at least 225 s after onset, at about the rate at which it revoked honest beacons. The behaviour-only detector isolated the A2 attacker in every seed and the A1 attacker in at most 4 of 20. Each literature family is blind to the attack outside its evidence class, as Table 1 predicts from published descriptions.

The disjunction of the two baselines is the more instructive comparison, and on it the

Table 10: Localisation error in metres, means across five seeds of the original campaign. "Attacker" covers receptions from the compromised beacon; "all" covers every monitored beacon, of which two of three are honest.

Scenario	Scope	RMSE	MAE	Median	P95
Clean	all	0.05	0.02	0.00	0.06
A1	attacker	267.25	265.42	271.41	299.23
	all	145.27	78.78	0.01	288.77
A2	attacker	0.03	0.01	0.00	0.05
	all	0.04	0.01	0.00	0.05
A3	attacker	146.02	145.81	145.91	150.01
	all	77.88	41.66	0.01	150.00

pipeline does not win. LND+WD covered every attack in every channel with false isolation statistically indistinguishable from the pipeline's: the paired difference in clean-network Φ (pipeline minus LND+WD) was 19 node-seconds at 0 dB (95% CI $[-22, 61]$) and -10 at 4 dB ($[-62, 43]$), and both were zero at 8 dB. It isolated attackers as fast or faster. The pipeline was slower by a mean of 3.7 s on A1 at 0 dB (CI $[2.0, 5.2]$), by 60 s on A2 at 4 dB ($[18, 104]$) and by 77 s on A3 at 8 dB ($[33, 122]$), and no paired interval showed it faster.

This is what Section 6.3 predicts. A1 and A2 each disturb one branch only, and A3 at its evaluated intensity lies above both branches' individual thresholds rather than in the fusion-only region, so for every attack evaluated here a disjunction loses nothing to fusion and avoids the attenuation of a live branch by a silent one. The latency gap is largest under shadowing, plausibly because the pipeline's behavioural branch also waits for its confidence weight to saturate, which a reputation threshold does not. In this network, therefore, the pipeline's detection is matched by two calibrated classical detectors run in parallel. Sections 6.14 and 6.15 show that this does not carry over to a sparse or a larger network, where the same disjunction pays several times the pipeline's false isolation.

6.7 Localisation error

Throughput is uninformative for A1, since the attack has no packet-level footprint. The service A1 degrades is localisation, so we report the displacement between the position a beacon advertises and its true position, which is the error a consumer of the localisation service inherits (Table 10).

The clean-network figure of 0.05 m RMSE confirms that honest beacons advertise their true positions to within numerical precision, so any error observed elsewhere is attributable to the attack. A2 produces 0.04 m, indistinguishable from clean: A2 is a pure forwarding attack with no positional signature, exactly as A1 is a pure positional attack with no forwarding signature. A3's attacker error of 146.02 m against A1's 267.25 m confirms the half-intensity construction, the ratio being 0.55 rather than 0.50 because clamping binds less often at the smaller offset.

Table 11: Isolation policy comparison in the enforced runs, five seeds per cell. Times in seconds, means over seeds; the attack begins at $t = 30$ s. Policy C performs no isolation.

Attack	Policy	Isolation time	R_{TH}	Recovery time
A1	C	—	0.97	—
	D	20.0	0.92	5.2
	E	45.0	0.95	44.8
A2	C	—	0.93	—
	D	20.0	0.94	6.2
	E	54.0	0.90	25.6
A3	C	—	0.93	—
	D	20.0	0.95	4.2
	E	45.0	0.94	35.0

The substantive result is the "all" row. Under A1 the mean error across all monitored beacons is 145.27 m while the median is 0.01 m. Two of three beacons are honest and contribute essentially zero error, while the third contributes the full displacement. A localisation consumer drawing on the affected beacon inherits an error more than three orders of magnitude above the clean-network floor, and averaging over honest neighbours does not conceal it, since trilateration depends on all contributing anchors.

6.8 Isolation policy under attack

Table 11 and Fig. 6 report the enforced runs of the original campaign, five seeds per cell, and Fig. 5 the time of first isolation under each policy over the 30-seed campaign. Detection is identical across policies within each attack: the policies differ only in what follows a verdict.

Policy D first isolates at a median of 15 s in every scenario, including the clean network, before the attack begins at 30 s. The 20.0 s in Table 11 is a five-seed mean (individual seeds 10–35 s, median 20 s), and is the same in all three attacks because the first isolation precedes the attack and so depends only on the seed. The enforced runs used the original campaign's calibration rather than the held-out seeds 101–105, which moves one of the five seeds from 20 to 35 s. What D removes at that point cannot be the attacker. Its apparently fast recovery time of 4.2–6.2 s is consequently an artefact of having isolated before there was anything to recover from. D_w and D_1 first isolate at medians of 30 and 35 s. Policy E first isolates at a median of 45 s under A1 and A3 and 57.5 s under A2, which is 30 s plus the corresponding detection latency: confirmation and enforcement coincide.

Recovery ratios span 0.90–0.97 across all nine cells, with confidence intervals substantially exceeding the differences between them (Fig. 6). We claim no throughput advantage for any policy. Throughput debt, which we had intended as a summary recovery metric, proved unusable: it returns non-zero values on clean runs, because the baseline window is

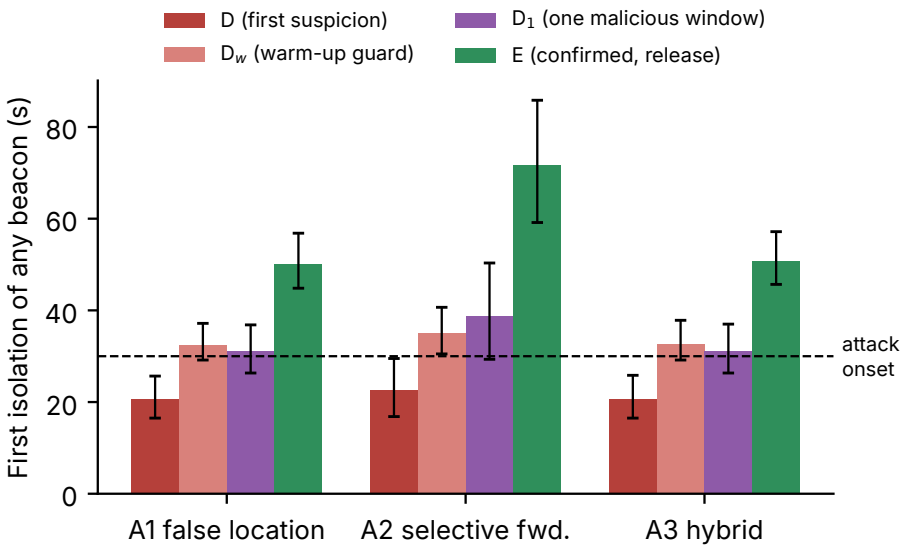


Figure 5: First isolation of any monitored beacon by policy, 30 seeds, with bootstrap 95% intervals. Policy D typically fires before the attack begins, so what it first removes cannot be the attacker; D_w and D_1 fire just after onset, and E fires at onset plus detection latency.

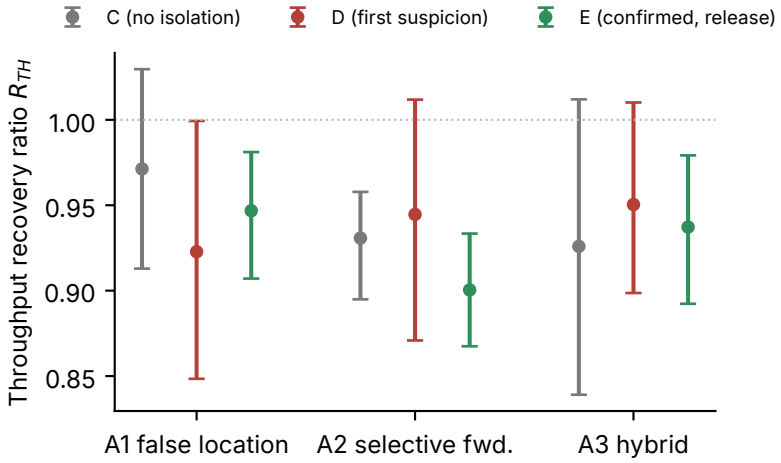


Figure 6: Throughput recovery ratio R_{TH} in the enforced runs, five seeds, with bootstrap 95% intervals. The intervals exceed every between-policy difference, so we claim no throughput advantage for any policy.

short relative to the throughput variance of a mobile topology. Correcting it requires a later attack start and a wider baseline window.

6.9 Mechanism cost

Fig. 7 decomposes what each policy does within a single run. The counters were logged per run but not aggregated across the campaign, so the figure illustrates the accounting without establishing a general magnitude. Both policies suppress attacker drops substantially

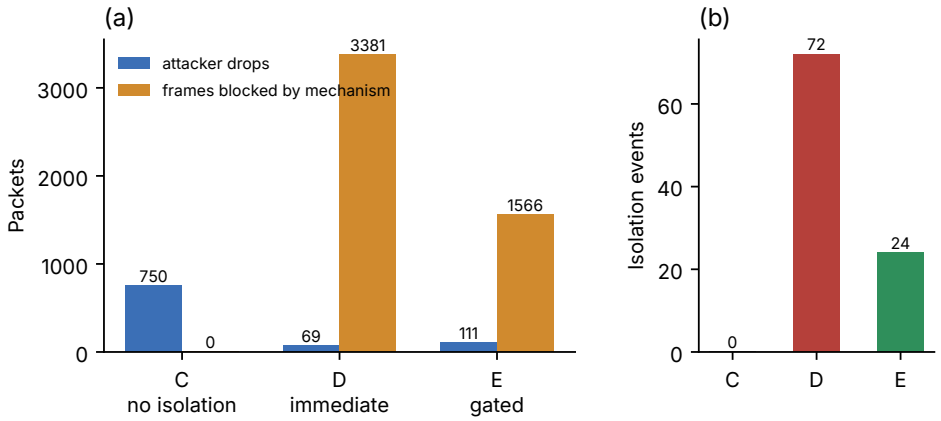


Figure 7: Isolation mechanism accounting in one illustrative enforced run (A2, seed 1), not an aggregate. Policy D suppresses more attacker drops but blocks more than twice as many frames and performs three times as many isolation events as policy E.

Table 12: Cumulative false isolation Φ (node-seconds) of honest beacons by policy, 30 seeds. Honest isolated counts node-seeds; 95% intervals are percentile bootstrap over seeds. The final column is the paired difference from policy E.

Network	Policy	Honest isolated	Φ	95% CI	$\Phi - \Phi_E$ [95% CI]
Clean	D	90/90	1052	[1025, 1077]	1005 [964, 1044]
	D_w	90/90	1018	[994, 1039]	971 [933, 1006]
	D_1	88/90	897	[843, 950]	851 [785, 911]
	E	9/90	47	[18, 79]	—
A1	D_1	59/60	611	[573, 648]	565 [519, 609]
	E	8/60	47	[18, 79]	—
A2	D_1	58/60	587	[532, 638]	557 [499, 612]
	E	7/60	30	[10, 54]	—
A3	D_1	59/60	586	[526, 639]	571 [514, 625]
	E	6/60	15	[2, 31]	—

relative to the control: from 750 packets under C to 69 under D and 111 under E. Frames blocked by the enforcement mechanism itself differ by more than a factor of two, 3381 under D against 1566 under E, tracking isolation-event counts of 72 and 24. The counter separates the mechanism's benefit from its cost within a run, which aggregate PDR does not.

6.10 Cumulative false isolation

The comparison that separates the policies is not made under attack. On a clean network every isolation is a false isolation, and Φ from (22) is directly measurable. Table 12 and Fig. 8 report it over 30 seeds for every policy, on the clean network and under each attack.

Policy D isolates on the first suspicious window, which typically falls before any attacker is active. To test whether the comparison rests on a start-up transient, D_w

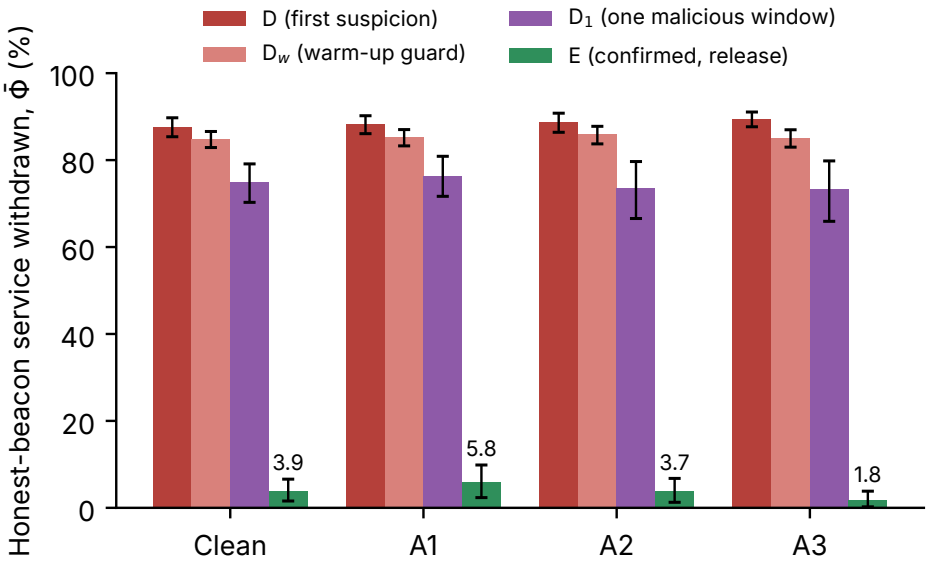


Figure 8: Fraction of honest-beacon service withdrawn, $\bar{\Phi}$, by policy and network, 30 seeds, with bootstrap 95% intervals. Every immediate policy withdraws 73–90% of honest service; confirmation-gated isolation (E) withdraws 2–6%.

ignores each beacon's first three usable windows, and D_1 isolates on the first window of malicious evidence without the persistence requirement. On the clean network D_w isolated all 90 honest node-seeds and D_1 88 of 90, with mean Φ of 1018 and 897 node-seconds against 47 for policy E, which isolated 9 of 90. The paired difference between D_1 and E is 851 node-seconds (bootstrap 95% CI [785, 911]); in normalised form D_1 withdraws 74.8% of honest-beacon service and E 3.9%. The advantage therefore does not come from the start-up transient: removing the persistence requirement alone is enough to isolate nearly every honest beacon. Under attack the same ordering holds, and every paired interval excludes zero. Replaying the original campaign's five clean seeds under that campaign's own calibration gives D 1054 and E 49 node-seconds, inside the corresponding 30-seed intervals. This is a check on calibration, not on the replay itself: both figures are computed offline.

The replay itself was tested on clean seeds 1–10 by enforcing the replayed schedules and rebuilding them from the verdicts issued under isolation (Section 5). Isolation lowered Φ for both policies: D_1 from 910.5 to 829.5 node-seconds (paired difference -81, 95% CI $[-201.5, 45.5]$) and E from 32.5 to 24.5 (difference -8, CI $[-22, 0]$). Neither interval excludes zero, and per seed the D_1 difference ranged from -370 to $+295$, so the replay is unbiased within this resolution but noisy for the immediate policy, and per-seed values of Φ for D_1 should not be read individually. The comparison the paper rests on survived: under isolation D_1 withdrew 69.1% of honest-beacon service and E 2.0%, D_1 exceeded E in all ten seeds, and the paired difference was 805 node-seconds (CI $[693.5, 917]$) against 878 in the replay. For E the replay, if anything, overstates Φ.

The mechanism is that the policies inherit different bounds. Policy E isolates only on confirmation, so its false-isolation rate is bounded by the detector's false-confirmation rate of about 10% (Section 6.4). The immediate policies isolate on weaker evidence, which on an uncontested network fires on nearly every monitored beacon. The three-state verdict's contribution is the tighter bound; it does not change what the detector can see. Section 6.11 shows that release contributes little to the gap.

6.11 Release under isolation

The schedules above were built from verdicts issued while no beacon was isolated. Re-verifying the enforced runs of the original campaign exposed a behaviour that this construction hides. While a beacon is isolated it forwards no transit traffic, so its behavioural branch falls below the minimum sample: across the enforced runs behavioural evidence was usable in 1.6% of the windows in which a beacon was isolated, against 54% for the same beacons and windows without isolation, while localisation evidence remained usable in every window. For the A2 attacker, whose only signature is behavioural, the fused trust therefore recovers once it is isolated. When the E schedule was rebuilt from the verdicts issued under isolation, the A2 attacker was isolated at 45–65 s and released at 115–160 s in all five seeds, after the twelve clean windows that clearing requires. A1 and A3 remained isolated, because their position lie keeps the localisation branch in its tail. In a closed loop E would therefore cycle a forwarding attacker between isolation and re-admission. The replayed schedules contain milder cases of the same effect: even without isolation, E released the attacker in 7 of the 90 attacked runs of the 30-seed campaign and in 3 of 60 at 4 dB.

Policy E′ (Section 4) releases a beacon confirmed on behavioural evidence only when that evidence returns, which an isolated beacon cannot supply. Its closed-loop schedule is therefore exact for those beacons, which stay isolated, and follows the replay for beacons confirmed on localisation alone. Table 13 gives its cost.

Evidence-aware release removed re-admission entirely: no attacker isolated under E′ was released, in any of the 209 runs in which one was isolated. The cost in false isolation was small. On the clean network $\bar{\Phi}$ rose from 3.9% to 5.0% (paired increase 13 node-seconds, 95% CI [2, 28]), and E′ withdrew at most 7.5% of honest-beacon service in any condition. The gap to D_1 remained between 372 and 837 node-seconds, with every interval excluding zero. The small cost has a revealing cause: 46 of the 50 honest beacons that E′ isolated across the three channels were never released, so E′ amounts to near-permanent isolation on confirmation. Its advantage over the immediate policies therefore comes from confirmation gating, not from release. Release under E affected few honest beacons, and on the evidence here a releasing policy is worth having only if it can distinguish recovered evidence from absent evidence.

Table 13: False isolation under evidence-aware release. $\bar{\Phi}$ is the fraction of honest-beacon service withdrawn; E′ is evaluated as the closed loop would run it. The last column is the paired difference $D_1 - E'$ in node-seconds with its bootstrap 95% interval. 30 seeds at 0 dB, 20 under shadowing.

σ_{sh}	Network	$\bar{\Phi}$ E	$\bar{\Phi}$ E′	$\bar{\Phi}$ D_1	$D_1 - E'$ [95% CI]
0 dB	Clean	0.039	0.050	0.748	837 [773, 901]
	A1	0.058	0.075	0.764	551 [500, 601]
	A2	0.037	0.047	0.734	550 [486, 607]
	A3	0.018	0.035	0.732	558 [497, 617]
4 dB	Clean	0.024	0.045	0.688	771 [677, 858]
	A1	0.016	0.029	0.694	532 [475, 586]
	A2	0.011	0.014	0.686	538 [482, 589]
	A3	0.008	0.008	0.658	520 [445, 584]
8 dB	Clean	0.000	0.000	0.602	723 [656, 786]
	A1	0.000	0.000	0.622	498 [452, 541]
	A2	0.036	0.036	0.514	382 [321, 449]
	A3	0.026	0.026	0.491	372 [298, 438]

Table 14: Aggregation ablation, original campaign. Confirmations out of 15 target-seeds per cell.

Configuration	A1	A2	A3	Clean
Direct only	6/15	5/15	6/15	1/15
Direct + indirect	6/15	5/15	6/15	1/15
Direct + indirect + historical	6/15	5/15	6/15	1/15

Counts include false confirmations: A1 6/15 is the attacker in five seeds plus one honest beacon.

Iterating the loop confirmed both findings. Over three rounds on ten seeds, E released the A2 attacker in every seed once its own enforced verdicts were used: the fraction of post-onset time the attacker spent isolated fell from 0.88 in the first round to 0.23 in the second, and recovered only to 0.78 in the third as the re-admitted attacker was confirmed again, the oscillation the mechanism predicts. Under E′ the attacker was never released in any round and spent 0.88 of the post-onset time isolated throughout. On the clean network E′ was a fixed point from the first round in all ten seeds, with Φ unchanged at 37 node-seconds, whereas E's Φ moved between rounds (32.5, 24.5, 33.5) in two of the ten.

The A2 throughput and recovery rows of Table 11 were produced by a schedule that never released the attacker. They are the rows E′ would produce, not those of E in a closed loop.

6.12 Aggregation ablation

Table 14 and Fig. 9 report the second stage ablated across its components. History-only is not an admissible configuration, since T^H seeds from the combined value. The three configurations are identical in every cell. The components are not inert in the underlying

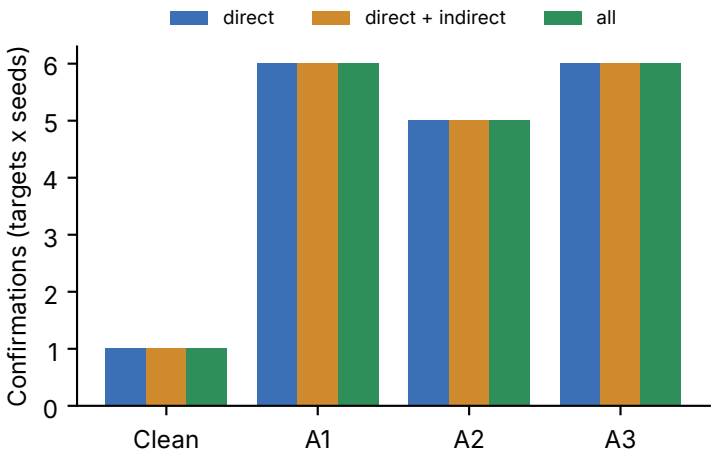


Figure 9: Aggregation ablation. The three configurations are identical in every cell: adding indirect and historical trust changes no verdict.

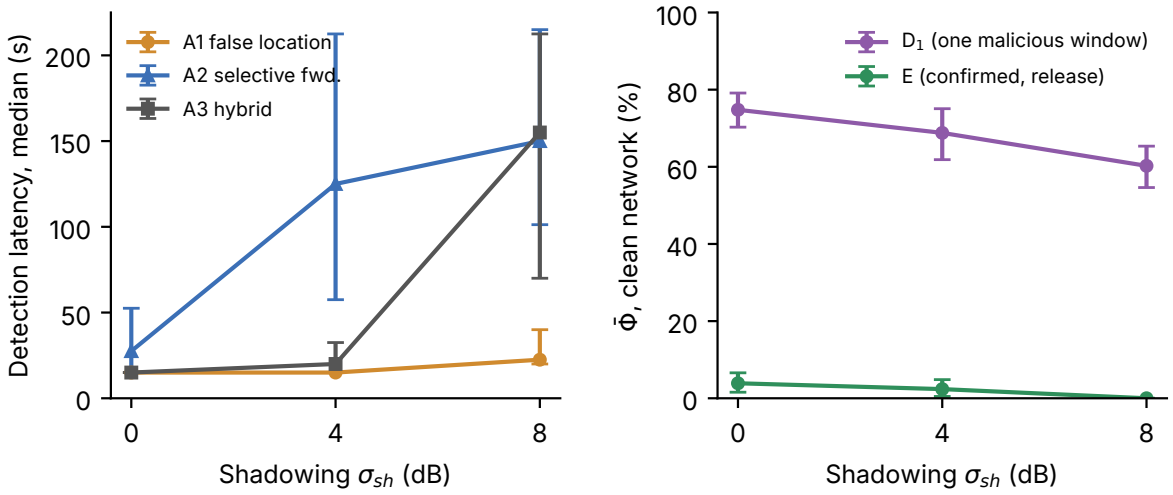


Figure 10: Effect of log-normal shadowing. Left: median detection latency, with interquartile range, rises sharply for attacks seen through forwarding behaviour. Right: false isolation on the clean network stays low for policy E and high for D_1.

computation: on A2 seed 1, direct-only aggregation yields mean trust 0.7148 with 50 window-level confirmations, while direct plus indirect yields 0.7548 with 70. But the differences never cross the persistence gate, so no target's final verdict changes. T^H is an exponentially weighted average of direct trust and T^I a trimmed mean of peers' direct trust: both are smoothed functions of the same underlying signal, and they would be expected to diverge only where direct trust is noisy and observers disagree, a regime our topology does not produce often enough to matter.

Table 15: Shadowing robustness, 20 seeds per cell. "Confirmed" is the fused pipeline's attacker detections; latency is the median [interquartile range] time from attack onset to confirmation, since the distributions are strongly right-skewed; $\bar{\Phi}$ is the fraction of honest-beacon service withdrawn. The $\sigma_{sh} = 0$ rows are the 30-seed campaign. In one 4 dB seed the beacon that later attacks was already falsely confirmed 5 s before onset; in the clean run of that seed the false confirmation cleared at 115 s, but in all three attack runs it persisted to the end, so the detection is counted. Its latency is undefined and that seed is excluded from the latency columns. It is also the single localisation confirmation of A2 at 4 dB.

σ_{sh}	Attack	Beh.	Loc.	Confirmed	Latency (s)	$\bar{\Phi}$ E / D_1
0 dB	A1	3/30	30/30	30/30	15.0 [15.0–15.0]	0.058 / 0.764
	A2	30/30	0/30	30/30	27.5 [15.0–52.5]	0.037 / 0.734
	A3	30/30	30/30	30/30	15.0 [15.0–15.0]	0.018 / 0.732
	Clean	—	—	—	—	0.039 / 0.748
4 dB	A1	5/20	20/20	20/20	15.0 [15.0–15.0]	0.016 / 0.694
	A2	20/20	1/20	20/20	125.0 [57.5–212.5]	0.011 / 0.686
	A3	20/20	20/20	20/20	20.0 [15.0–32.5]	0.008 / 0.658
	Clean	—	—	—	—	0.024 / 0.688
8 dB	A1	0/20	20/20	20/20	22.5 [20.0–40.0]	0.000 / 0.622
	A2	20/20	3/20	20/20	150.0 [101.2–215.0]	0.036 / 0.514
	A3	14/20	18/20	19/20	155.0 [70.0–212.5]	0.026 / 0.491
	Clean	—	—	—	—	0.000 / 0.602

6.13 Robustness to shadowing

The campaign above uses a channel without shadowing, which flatters RSSI ranging. We repeated it under log-normal shadowing with $\sigma_{sh} = 4$ and 8 dB, 20 seeds per attack and channel. The pipeline was recalibrated on clean runs of each channel, as a deployment would be; no threshold was changed. Table 15 and Fig. 10 give the result.

Detection survived. The fused pipeline confirmed the attacker in every seed at 4 dB and in 59 of 60 at 8 dB, the miss being one A3 seed. The coverage pattern held: localisation alone confirmed the A2 attacker in at most 3 of 20 seeds, and behaviour alone confirmed the A1 attacker in 5 of 20 at 4 dB and none at 8 dB. Behavioural evidence degraded first: at 8 dB it confirmed the A3 attacker in 14 of 20 seeds, against 20 of 20 at 4 dB. Shadowing adds link loss that honest forwarders cannot control, which widens the peer distribution the behavioural score is measured against.

The cost appeared as latency. Median time to confirmation for A2, the attack visible only through forwarding, rose from 27.5 s without shadowing to 125 s at 4 dB and 150 s at 8 dB; A3 rose to 155 s at 8 dB, once its behavioural branch had degraded. A1, with its strong localisation signature, was confirmed at a median of 15 s at 4 dB and 22.5 s at 8 dB.

False isolation did not grow. Policy E isolated at most 5 of 40 honest node-seeds under attack and withdrew at most 3.6% of honest-beacon service. Recalibration widened the clean-network null enough that E isolated no honest beacon in any clean 8 dB seed. The

Table 16: Sparse topology (25 nodes, 1000×1000 m, 10 seeds). Detection counts attackers isolated after onset (for the pipeline, confirmed); latency is the median in seconds; FI and $\bar{\Phi}$ are for the clean network (E′ for the pipeline).

Method	A1	A2	A3	FI	$\bar{\Phi}$
Pipeline	10/10	4/10	10/10	0/30	0.000
latency	62.5	132.5	62.5		
LND	10/10	0/10	10/10	0/30	0.000
WD	8/10	9/10	9/10	16/30	0.239
LND+WD	10/10	9/10	10/10	16/30	0.239
Policy D_1				8/30	0.100
Policy D				30/30	0.849

immediate policies also isolated less as the channel worsened, because the same widening raised their effective threshold, but D_1 still withdrew 60–69% of honest-beacon service on clean networks, and the bootstrap intervals of D_1 and E did not overlap in any condition.

6.14 Sparse topology

Spreading the same 25 nodes over a 1000×1000 m field changed the balance of evidence. Behavioural evidence was usable in 18% of windows, against 52% in the dense topology, while localisation evidence remained usable in every window. Table 16 gives the result.

The pipeline isolated no honest beacon in any sparse run, but it confirmed the forwarding-only attacker in only 4 of 10 seeds and took a median of 62.5 s to confirm the attacks with a localisation signature, four times the dense figure. The cause is evidence scarcity: with few transit packets per window the behavioural branch rarely reaches its minimum sample and its confidence weight stays low. The behaviour-only reputation baseline, which has no confidence weighting, isolated the A2 attacker in 9 of 10 seeds, but on the clean network it also isolated 16 of 30 honest node-seeds and withdrew 23.9% of honest-beacon service, because broken routes in a sparse network look like dropped packets. The disjunction inherited both properties. The paired difference in clean-network Φ between the pipeline and LND+WD was -287 node-seconds (95% CI $[-446, -149]$), and the pipeline's Φ was lower under every attack as well (CIs excluding zero).

The comparison of Section 6.6 therefore depends on the topology. In the small dense network the disjunction matched the pipeline's false isolation and was faster; in the sparse one it detected the forwarding attacker more often but at roughly a quarter of honest-beacon service, which is the trade Φ was introduced to expose. One caveat applies to the sparse calibration: every usable honest behavioural window in the calibration runs forwarded all its transit packets, so the robust scale of the behavioural statistic collapsed to its floor. That made the branch maximally sensitive whenever it had evidence, which reinforces rather than explains its low detection rate, but a calibration with more

Table 17: 50-node network, two attackers, six monitored beacons, 10 seeds. Detection counts runs in which both attackers were isolated after onset; latency is the median over runs of the mean time to confirm the two attackers, in seconds; FI and $\bar{\Phi}$ are for the clean network (E′ for the pipeline).

Method	A1	A2	A3	Latency A1 / A2 / A3	FI	$\bar{\Phi}$
Pipeline	10/10	10/10	10/10	15 / 27.5 / 15	5/60	0.026
LND	10/10	0/10	10/10	11.3 / — / 17.5	2/60	0.005
WD	0/10	9/10	9/10	— / 20 / 67.5	7/60	0.108
LND+WD	9/10	9/10	9/10	12.5 / 20 / 15	9/60	0.113
Policy D_1					60/60	0.742
Policy D					60/60	0.901

behavioural samples is needed before the sparse behavioural figures are relied on.

6.15 A larger network with two attackers

Doubling the network to 50 nodes at the same density, with six monitored beacons of which two attack simultaneously, left the main results in place and sharpened the baseline comparison. Table 17 gives the result.

The pipeline confirmed both attackers in every attacked run, with medians of 15–27.5 s, and isolated 5 of 60 honest node-seeds on the clean network. Each single-evidence baseline was again blind outside its evidence class: the localisation-only detector isolated no A2 attacker pair and the behaviour-only detector no A1 pair. The coverage result therefore holds with two simultaneous attackers and at twice the scale.

The disjunction of the baselines no longer matched the pipeline. It missed one attacker in one run of each attack scenario, and its behavioural half made it withdraw about four times as much honest-beacon service: 11.3% on the clean network against the pipeline's 2.6%. Paired across seeds, the pipeline's Φ was lower by 154 node-seconds under A2 (95% CI $[30, 277]$) and by 155 under A3 ($[42, 276]$); under A1 (-163, $[-342, 8]$) and on the clean network (-209, $[-458, 0]$) the intervals reached zero. Detection speed was indistinguishable: no paired latency difference excluded zero. Immediate isolation was as costly as in the small network, with D_1 withdrawing 72–74% of honest-beacon service, and the gap to E′ exceeded 1100 node-seconds under every scenario.

Taken with Section 6.14, this bounds the baseline comparison of Section 6.6. Only in the small dense network did two classical detectors run side by side match the fused pipeline. In the sparse network and in the larger one, the behaviour-only reputation detector, which has no confidence weighting or peer-relative scoring, paid for its sensitivity in false isolation, and the pipeline's lower Φ is the measured benefit of combining the evidence.

7 Discussion

7.1 Interpretation

The coverage result is structural, following from what each evidence class can physically observe rather than from a threshold setting, and it was obtained with calibration held fixed across all scenarios. The implication for the literature is narrow but practical. An accuracy reported against localisation-visible attacks is an accuracy over that subset of the attack space. EMBN-MANET's threat model is confined to false-location behaviour by design, so its figure is correct within its scope and we do not dispute it. What we would ask of any such report is one additional line: which attack classes the evidence can observe at all.

The isolation result is the paper's strongest. Over 30 seeds, confirmation-gated isolation withdrew 3.9% of honest-beacon service on an uncontested network (5.0% with evidence-aware release) against 74.8% for the strongest immediate policy, with every paired interval excluding zero; the ordering held under attack, under shadowing and when the schedules were enforced. With evidence-aware release, its cost is the time an attacker remains active before confirmation: a median of 15 s for attacks with a localisation signature and 27.5 s for selective forwarding, rising to medians of 125–155 s for forwarding attacks under shadowing. Whether that trade is favourable depends on the prior probability of attack and on what a wrongly removed beacon costs the network. Where attack is rare and beacon coverage sparse, gating is preferable, since Φ accumulates continuously while attacker exposure is bounded by the confirmation delay. That bound holds only if release cannot be triggered by missing evidence: under plain E a forwarding attacker is re-admitted once its isolation silences its behavioural branch. Release is as consequential a decision as isolation, and in our data the safe release rule made isolation on confirmation close to permanent. Where attack is frequent and beacon redundancy high, immediate isolation may be preferable. We present the trade rather than a winner, and suggest that Φ or a comparable measure be reported routinely: a mechanism's false-positive rate says nothing about how long its errors persist.

7.2 Limits of the evidence

Whether fusion helped was decided by the attack family, not by the combination rule. In the primary sweep the branches' sensitivity ratio was 3.63, beyond the bound of 2.414, and Stouffer combination confirmed fewer attackers than a disjunctive rule at the two intensities nearest the behavioural threshold. Halving the drop rate relative to displacement brought the ratio to 1.87, inside the bound, and fusion then never confirmed fewer and confirmed one more. The detector was the same in both sweeps; only the mix of displacement and drop rate changed. Fusion contributes against attackers whose two

misbehaviours are comparably visible to the two branches, and against lopsided ones a disjunctive rule does as well or better. The baseline comparison is the same result at the level of whole detectors: in the small dense network, two calibrated single-evidence detectors run in parallel matched the fused pipeline on every attack evaluated and were faster under shadowing. In the sparse and 50-node networks the behaviour-only detector's false isolation rose to 11–24% of honest-beacon service while the pipeline's stayed at 0–3%, plausibly because of its confidence weighting and peer-relative scoring; there the benefit of fusion appears in Φ, not in detection. Branch sensitivity to a given attack family can be measured before fusion is applied, and no system we surveyed reports checking it. An adaptive adversary could exploit this by concentrating its misbehaviour on one axis, which is a further reason to keep a single-branch path to confirmation alongside the fused statistic.

The aggregation stage changed no verdict anywhere. It belongs in the framework as an unevaluated robustness extension, not as a validated component: a system using only direct per-observer trust would reproduce every reported finding. It remains because our campaign barely samples the conditions it exists to handle, namely observers who disagree, direct trust that is noisy, and recommenders who lie.

Neither isolation policy produced a defensible throughput benefit. This may be a property of a dense 25-node topology in which AODV finds alternative routes readily; enforcing isolation in the sparse topology is the obvious next experiment. Multipath recovery schemes [10] and isolation studies in related ad hoc settings [29, 38] suggest the effect may be topology-dependent. On the evidence we have, the claim that isolation restores network performance is unsupported. Finally, throughput-based evaluation of A1 measures nothing, since A1 has no packet-level footprint; the appropriate metric for that attack class is localisation error.

8 Threats to Validity

Construct validity. Behavioural monitoring is node-global rather than per-observer, so indirect trust operates on a coarser behavioural signal than the model specifies. This weakens the aggregation ablation: the null may reflect the implementation rather than the design. Φ for the 30-seed campaign is computed by applying each policy offline to the same verdict stream, which assumes that an isolation does not change later verdicts. For the never-releasing policies the first isolation time is exact and only later isolations in the same run can be affected; for E the release times can be, and Section 6.11 shows that for forwarding attackers they are. Enforcing the replayed schedules on ten clean seeds and rebuilding them from the verdicts issued under isolation moved neither policy's Φ by an amount distinguishable from zero and preserved the D_1–E gap in every seed (Section 6.10). Iterating the loop for three rounds left E′ at a fixed point on every clean

seed and never released an attacker, while E re-admitted the A2 attacker in every seed. Iteration approximates an online detector but is not one, and it covers E and E′ on two scenarios. E′ was evaluated from verdicts issued before isolation; its schedule is exact for beacons confirmed on behaviour, which it never releases, and inherits the replay assumption for beacons confirmed on localisation alone. E′ was designed after the enforced runs were examined, so its evaluation is not blind to that finding.

Internal validity. The recovery baseline window is short relative to throughput variance, which is why we exclude throughput debt. Detection thresholds were calibrated on held-out clean runs and fixed, which avoids tuning to the test scenarios but means no scenario received its optimal operating point.

External validity. The primary channel has no shadowing, multipath or obstruction model; the shadowing campaign shows that detection and the false-isolation result survive 4–8 dB, at a substantial cost in latency for forwarding attacks. The campaign is otherwise narrow in five ways. Attackers are non-colluding and non-adaptive. Mobility follows a synthetic model rather than a trace. The topologies are a 25-node network in a dense and a sparse field and a 50-node network at the dense field's density; the sparse and 50-node campaigns measured detection and false isolation but not the throughput effect of enforced isolation, so the throughput nulls rest on the small dense field alone. Larger networks were limited by memory: a 50-node run needs about 3 GB, and a 100-node configuration did not fit the hardware available. Three monitored beacons per run, six at 50 nodes, make rate-based metrics coarse. And only the hybrid attack's intensity is swept. We do not claim the results generalise beyond this configuration.

Comparative validity. The two literature baselines re-implement the evidence class and decision rule of Liu–Ning–Du alert counting and of watchdog/CONFIDANT reputation; they are not complete reimplementations of those systems, and they use our simulator's node-global behavioural counters and our residual. Their thresholds were set on calibration seeds by rules we chose, and a different but equally defensible rule would move their operating points. Policies D, D_w and D_1 are our own constructions, reasonable stand-ins for isolate-on-suspicion behaviour. The comparison against EMBN-MANET itself remains analytical, resting on a reading of its evidence class and threat model.

Statistical conclusion validity. Coverage, latency, false positives and Φ rest on 30 seeds with held-out calibration and paired bootstrap intervals. The throughput, recovery, localisation-error and ablation results rest on the five seeds of the original campaign; no difference among the continuous throughput measures is claimed. The sparse and 50-node campaigns use ten seeds each, so their paired intervals are wide and two of the 50-node comparisons reach zero. The sweeps use ten seeds per level, so the one-seed fusion gain in the second pairing is reported as consistent with the condition, not as a measured benefit.

9 Conclusion

We evaluated multi-evidence verification and confirmation-gated isolation of malicious beacon nodes in a mobile ad hoc network across 780 ns-3 simulation runs.

For the two attack classes evaluated, single-evidence detection is blind to attacks outside its evidence class, and collecting both behavioural and localisation evidence removes the observed blind spots. Confirmation-gated isolation reduced cumulative false isolation of honest beacons to 3.9% of available service against 74.8% for the strongest immediate-isolation policy over 30 seeds, at a median cost of 15–27.5 s of attacker activity after onset, and the gap held when the schedules were enforced. Releasing a beacon when its verdict clears re-admitted forwarding attackers, whose behavioural evidence disappears while they are isolated; requiring that evidence to return before release prevented every re-admission and raised false isolation only to 5.0%. Both results held under 4 and 8 dB shadowing, where the price moved to latency: forwarding attacks took a median of 125–155 s to confirm. Every weighted Stouffer detector has a fusion-only region, but an attack family scaled in intensity reaches it only when the branches' detection thresholds for that family lie within a factor of $1 + \sqrt{2}$, and across two sweeps that condition predicted whether fusion diluted ($r = 3.63$) or matched and marginally exceeded ($r = 1.87$) a disjunctive rule. Consistent with that condition, a calibrated disjunction of localisation-only and behaviour-only literature baselines matched the fused pipeline's coverage and false isolation and isolated attackers as fast or faster in the small dense network. In the sparse network the disjunction detected the forwarding attacker more often but withdrew 24% of honest-beacon service, against none for the pipeline, and in a 50-node network with two attackers it withdrew 11% against 3% while the pipeline confirmed every attacker. The trust aggregation layer changed no verdict, and neither isolation policy produced a throughput difference exceeding its confidence interval.

We report these outcomes together because the negative ones bound the positive ones. In a literature where reported accuracies cluster above 95%, the more informative questions are which attacks a detector can see and how long its mistakes last.

9.1 Future work

Complete baselines. Full implementations of EMBN-MANET and of a behavioural trust-routing scheme, including their response mechanisms, on the same topology and traces.

Scale and topology. Networks of 100 nodes and more, enforced isolation in sparse topologies where it can affect delivery, varied mobility speeds, and colluding or adaptive attackers.

Fusion-rule comparison. Bounded rules (arithmetic and geometric mean), calibrated statistical rules (Stouffer, Fisher, Dempster–Shafer) and learned rules (regularised

logistic regression, gradient-boosted trees) compared on identical oriented evidence, in configurations chosen on both sides of the bound (28), with folds partitioned by seed.

Aggregation in its intended regime. Noisy observers, conflicting observers and malicious recommenders, which is where indirect and historical trust are designed to matter.

Post-isolation localisation error. Splitting localisation error at the isolation timestamp would give direct evidence that isolation restores a degraded service, which throughput cannot show for A1.

Online closed-loop evaluation. The policies here were evaluated by replay, one-step enforcement and a three-round iterated loop. Running the detector inside the simulator, so that each isolation and release feeds back into later verdicts, would test E' and the replay assumption directly, and would allow probationary release, in which a released beacon is re-isolated on its first malicious window.

Faster behavioural confirmation under shadowing. The latency cost of forwarding attacks under shadowing comes from the behavioural branch's slow confidence saturation and widened peer distribution; adaptive half-weights or a second behavioural observable could reduce it.

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Declarations

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Author contributions. A.V. designed and implemented the simulator, verification pipeline and experiments, and drafted the manuscript. D.V.C. supervised the work, contributed to the experimental design and revised the manuscript. Both authors read and approved the final version.

Data and code availability. Simulation sources, the analysis pipeline, the scripts that run every campaign and regenerate every data figure, and the result tables, per-window verdicts and isolation schedules for every campaign are available at `https://github.com/kraghu356/manet-beacon-trust`. The raw per-run simulator traces are available from the corresponding author on reasonable request. No third-party dataset was used: every data point in this paper was generated by the authors' simulations.

Ethics approval. Not applicable. This study involves neither human participants nor animals.

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